Uncertainty Quantification for Large Language Models

Artem Shelmanov, Maxim Panov,
Roman Vashurin, Artem Vazhentsev, Ekaterina Fadeeva, Lyudmila Rvanova,
Timothy Baldwin

The 40th Annual AAAI Conference on Artificial Intelligence
(AAAI-2026)

January 20, 2026



Mohamed bin Zayed
University of
Artificial Intelligence

ETH zürich



1. Introduction

Artem Shelmanov

Who we are



Artem Shelmanov
Senior Research Scientist
MBZUAI



Maxim Panov
Assistant Professor
MBZUAI



Timothy Baldwin
Professor, Provost
MBZUAI



Roman Vashurin
Research Engineer
MBZUAI



Artem Vazhentsev
Postdoc Researcher
MBZUAI



Lyudmila Rvanova
NLP Researcher
Independent



Ekaterina Fadeeva
PhD student
ETH Zürich

Materials and Q&A

Tutorial materials



Ask your questions



After every major section there will be a 5-minute break for Q&A

LLMs hallucinate I

A Tell me a bio of Dr. Artem Shelmanov

AI Dr. Artem Shelmanov is a prominent figure in the fields of **neuroscience**, **artificial intelligence (AI)**, and **cognitive science**. While specific details of his personal life and early years are not widely publicized, his professional contributions and impact are well-documented. Here's a concise biography based on available information:

Early Life and Education

Artem Shelmanov demonstrated a keen interest in science and technology from a young age. He pursued higher education in **neuroscience** and **computer science**, likely earning advanced degrees (Master's and Ph.D.) from prestigious institutions. His interdisciplinary background allowed him to bridge the gap between biological and artificial intelligence systems.

Career and Research

Dr. Shelmanov is best known for his groundbreaking work in **neuro-AI interfaces**, where he explores how the human brain interacts with AI systems. His research focuses on:

- **Brain-Computer Interfaces (BCIs):** Developing technologies that enable direct communication between the brain and external devices.

LLMs hallucinate I

A Tell me a bio of Dr. Artem Shelmanov

AI Dr. Artem Shelmanov is a prominent figure in the fields of **neuroscience**, **artificial intelligence (AI)**, and **cognitive science**. While specific details of his personal life and early years are not widely publicized, his professional contributions and impact are well-documented. Here's a concise biography based on available information:

Early Life and Education

Artem Shelmanov demonstrated a keen interest in science and technology from a young age. He pursued higher education in **neuroscience** and **computer science**, likely earning advanced degrees (Master's and Ph.D.) from prestigious institutions. His interdisciplinary background allowed him to bridge the gap between biological and artificial intelligence systems.

Career and Research

Dr. Shelmanov is best known for his groundbreaking work in **neuro-AI interfaces**, where he explores how the human brain interacts with AI systems. His research focuses on:

- **Brain-Computer Interfaces (BCIs):** Developing technologies that enable direct communication between the brain and external devices.

LLMs hallucinate II

User: How many letters in the word "nineteen"?

LLM: The word "nineteen" has 9 letters.

User: No there are 12 letters in the word "nineteen."

LLM: My apologies, you are correct! The word "nineteen" does indeed have 12 letters: N-I-N-E-T-E-E-N. Thank you for pointing out my mistake.

LLMs hallucinate II

User: How many letters in the word "nineteen"?

LLM: The word "nineteen" has 9 letters.

User: No there are 12 letters in the word "nineteen."

LLM: My apologies, you are correct! The word "nineteen" does indeed have 12 letters: N-I-N-E-T-E-E-N. Thank you for pointing out my mistake.

Uncertainty as a universal hallucination detector

Consider: $\mathbf{x}$ is an input sequence; $\mathbf{y}$ is an output sequence of length L.

Autoregressive LLM models:

$$P(\mathbf{y} \mid \mathbf{x}) = \prod_{l=1}^{L} P(y_l \mid \mathbf{y}_{<l}, \mathbf{x}).$$

In addition to $P(\mathbf{y} \mid \mathbf{x})$, we would like to obtain an uncertainty score:

$$\mathrm{U}(\mathbf{x}, \mathbf{y}).$$

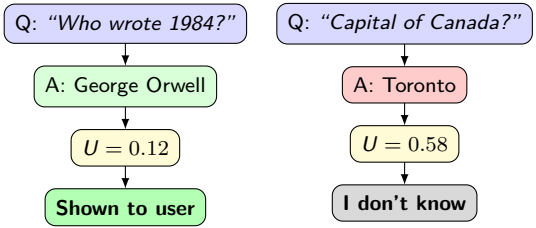
We want $\mathrm{U}(\mathbf{x}, \mathbf{y})$ to correlate with LLM mistakes and help us to solve **reliability tasks**, such as **selective generation** or **hallucination detection**

Reliability task: Selective generation

- **Goal:** Deliver answers only when the LLM is confident.
- **Selective rule:**

$$\text{output} = \begin{cases} \text{answer}, & \mathrm{U}(\mathbf{x}, \mathbf{y}) < \tau, \\ \text{“I don't know”}, & \mathrm{U}(\mathbf{x}, \mathbf{y}) \geq \tau. \end{cases}$$

- By **rejecting** uncertain answers, we increase performance for remaining answers.



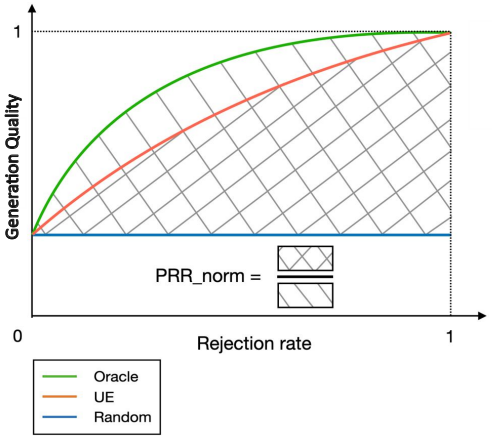
Example: with $\tau = 0.3$ we keep **70%** of answers while boosting accuracy from **80%** to **95%**

Important: not limited to question answering, also applies to machine translation, summarization, reasoning, ...

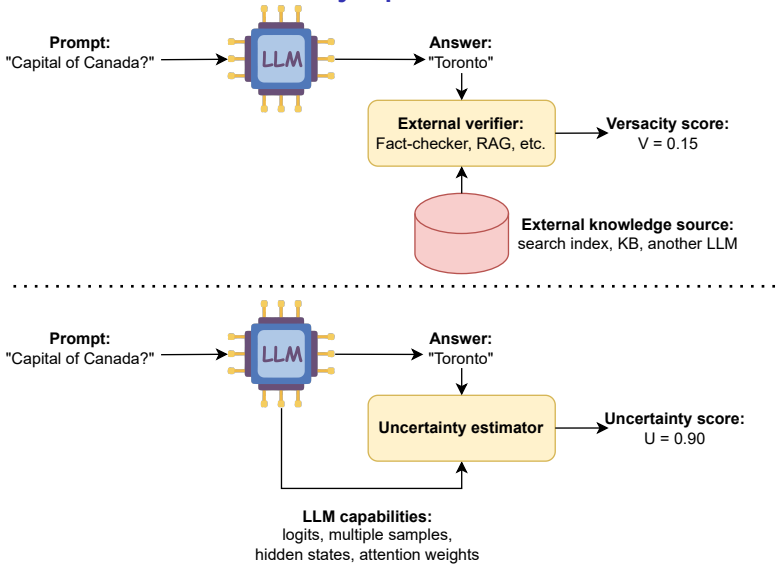
Rejecting uncertain predictions improves performance for remaining predictions

Prediction–rejection curve at rejection rate τ:

- Reject top-τ% most uncertain outputs
- Plot performance for remaining predictions



External verifier vs uncertainty quantification



Reliability task: Claim-level hallucination detection



Tell me a bio of Madonna.



Madonna is an American singer, songwriter, actress, and businesswoman. She was born on August 16, 1958, in Bay City, Michigan, and is best known for her provocative image and music. Madonna has sold over 300 million records worldwide, making her one of the best-selling music artists of all time. She has won numerous awards, including 20 Grammy Awards, a Golden Globe Award, and an Emmy Award. Madonna is also known for her philanthropic work and her activism on various social issues.

If $\mathrm{U}(\mathbf{x}, \mathbf{y}, c) \geq \tau$, highlight claim $c \in C(\mathbf{y})$ of LLM as potentially erroneous.

Other applications of uncertainty

- **Selective classification / generation**
- **Hallucination detection in LLM generations**
- **Improving performance and efficiency of multi-step reasoning systems: test time scaling and agentic AI**
- Out-of-distribution detection
- Active learning
- Adversarial attack detection
- Reinforcement learning / control theory

Goals & disclaimers ⚠

- **Main goals:**
  - Describe major ideas behind SOTA UQ methods for LLMs
  - Provide practical suggestions for deploying UQ in LLM-based applications
  - Facilitate further research in LLM safety by sharing the benchmarks, standard practices, and gaps in current research
- **Knowledge cutoff:** November 2025, focus on published works
- **Disclaimers:**
  - This tutorial is more practically oriented than theoretical
  - The tutorial is by no means exhaustive: we try to cover ideas, but not all papers
  - Biased by our own research experience
  - Email us, if you think that something crucial is missing:
    `artem.shelmanov@mbzuai.ac.ae`
    `maxim.panov@mbzuai.ac.ae`

Outline

1. **Introduction & LM-Polygraph** (15 mins)
2. **Theoretical background** (20 mins)
3. **Unsupervised UQ:** Information-based, consistency-based, and verbalized uncertainty (35 mins)
4. **Introspective and supervised UQ** (25 mins)

   [Break 30 minutes]
5. **UQ for LVLMs** (15 mins)
6. **Uncertainty normalization & calibration** (10 mins)
7. **Benchmarking UQ methods** (15 mins)
8. **Uncertainty for reasoning and agentic AI** (20 mins)
9. **Conclusion & future work** (10 mins)

What is left out of scope

- Bayesian deep learning
- Conformal prediction
- Calibration of LLM probabilities
- Fine-tuning LLMs for abstention and confidence expression

Previous related tutorials & resources

- **Tutorials:**
  - **Uncertainty quantification for LLMs.**
    Artem Shelmanov, Maxim Panov et al. ACL-2025
  - **Uncertainty Estimation for Natural Language Processing.**
    Adam Fisch, Robin Jia, Tal Schuster. COLING-2022
  - **Practical Uncertainty Estimation and OOD Robustness in Deep Learning**.
    Dustin Tran, Balaji Lakshminarayanan, Jasper Snoek. NeurIPS-2020
  - **Bayesian Deep Learning and a Probabilistic Perspective of Model Construction**. Andrew Gordon Wilson. ICML-2020

Previous related tutorials & resources

- **Tutorials:**
  - **Uncertainty quantification for LLMs.**
    Artem Shelmanov, Maxim Panov et al. ACL-2025
  - **Uncertainty Estimation for Natural Language Processing.**
    Adam Fisch, Robin Jia, Tal Schuster. COLING-2022
  - **Practical Uncertainty Estimation and OOD Robustness in Deep Learning**.
    Dustin Tran, Balaji Lakshminarayanan, Jasper Snoek. NeurIPS-2020
  - **Bayesian Deep Learning and a Probabilistic Perspective of Model Construction**. Andrew Gordon Wilson. ICML-2020
- **Workshops**:
  - **UncertaiNLP** @ EACL-2024, EMNLP-2025, [EMNLP-2026]
  - **QUESTION / Agentic AI in the Wild** @ ICLR-2025, @ ICLR-2026

Previous related tutorials & resources

- **Tutorials:**
  - **Uncertainty quantification for LLMs.**
    Artem Shelmanov, Maxim Panov et al. ACL-2025
  - **Uncertainty Estimation for Natural Language Processing.**
    Adam Fisch, Robin Jia, Tal Schuster. COLING-2022
  - **Practical Uncertainty Estimation and OOD Robustness in Deep Learning**.
    Dustin Tran, Balaji Lakshminarayanan, Jasper Snoek. NeurIPS-2020
  - **Bayesian Deep Learning and a Probabilistic Perspective of Model Construction**. Andrew Gordon Wilson. ICML-2020
- **Workshops**:
  - **UncertaiNLP** @ EACL-2024, EMNLP-2025, [EMNLP-2026]
  - **QUESTION / Agentic AI in the Wild** @ ICLR-2025, @ ICLR-2026
- **Benchmark: Benchmarking Uncertainty Quantification Methods for Large Language Models with LM-Polygraph.** Vashurin et al. @ TACL-2025

LM-Polygraph: Uncertainty quantification framework for LLMs

- Python library: **40+ SOTA UQ methods** with unified program API
- Built-in routines for **normalization / calibration**
- Works with both **textual & visual** LLMs
- Integrates with
  - Open-weight LLMs from **HuggingFace**: Llama, Gemma, Qwen, DeepSeek, etc.
  - Models deployed via **vLLM**
  - **LLMs as a service**: ChatGPT, Gemini, OpenRouter, etc.
- **Extendable benchmark** covering MT, QA, summarization, language understanding, fact-checking, reasoning

GitHub:

License:

LM-Polygraph: High-level code example

```
from lm_polygraph import estimate_uncertainty
from lm_polygraph.model_adapters import WhiteboxModel
from lm_polygraph.estimators import *

model = WhiteboxModel.from_pretrained(
    "meta-llama/Llama-3.1-8B-Instruct",
    device_map="cuda:0"
)

ue_method  = MeanTokenEntropy()
input_text = "Who is George Bush?"
estimate_uncertainty(model, ue_method, input_text)
```

LM-Polygraph: Low-level code example

```
# 1. Specify model adapter
# 2. Specify stat calculators -- processors that generate text from LLM and
#    outputs necessary for UQ methods
# 3. Specify uncertainty estimator
data_loader = DataLoader(chat_messages, batch_size=batch_size, ...)
for batch in data_loader:
    deps = dict()

    # Generate text from LLM and other outputs
    deps.update(stat_calc_1_LLM_generate(deps, texts=batch, model=model_adapter))
    deps.update(stat_calc_2_comp_sim_matrix(deps, texts=batch, model=model_adapter))
    ...
    deps.update(stat_calc_n(deps, texts=batch, model=model_adapter))

    uncertainty_scores = uncertainty_estimator(deps)
    generated_texts = tokenizer.batch_decode(deps['greedy_tokens'])
```

The core team behind LM-Polygraph

- Ekaterina Fadeeva
- Artem Vazhentsev
- Roman Vashurin
- Lyudmila Rvanova
- Akim Tsvigun
- Daniil Vasilyev
- Alexander Panchenko
- Preslav Nakov
- Timothy Baldwin
- Maxim Panov
- Artem Shelmanov



Mohamed bin Zayed University of Artificial Intelligence

ETH zürich





Takeaways & QA

- Uncertainty quantification poses a universal approach to hallucination detection
- Uncertainty quantification for generation tasks is more challenging than for classification and regression tasks
- LM-Polygraph provides implementations of SOTA UQ methods in Python that could be easily integrated into the LLM inference code
- LM-Polygraph provides a benchmark for research in uncertainty quantification

2. Theoretical Background

Maxim Panov

Uncertainty in classification

In binary classification, the value of predicted probability is a confidence in the positive class:

$$\widehat{p}(\mathbf{x}) = p\left(\mathbf{y} = 1 \mid \mathbf{x}\right).$$

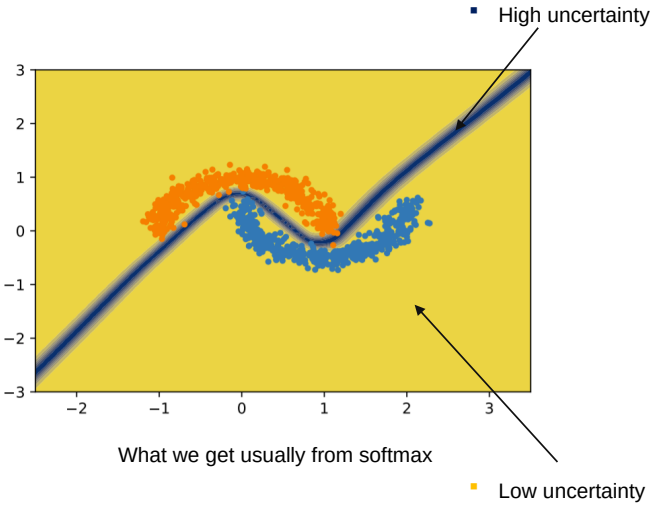
It generalizes to multiclass problems by using the value of the class with maximum probability:

$$\widehat{p}(\mathbf{x}) = \max_{c} p\left(\mathbf{y} = c \mid \mathbf{x}\right).$$

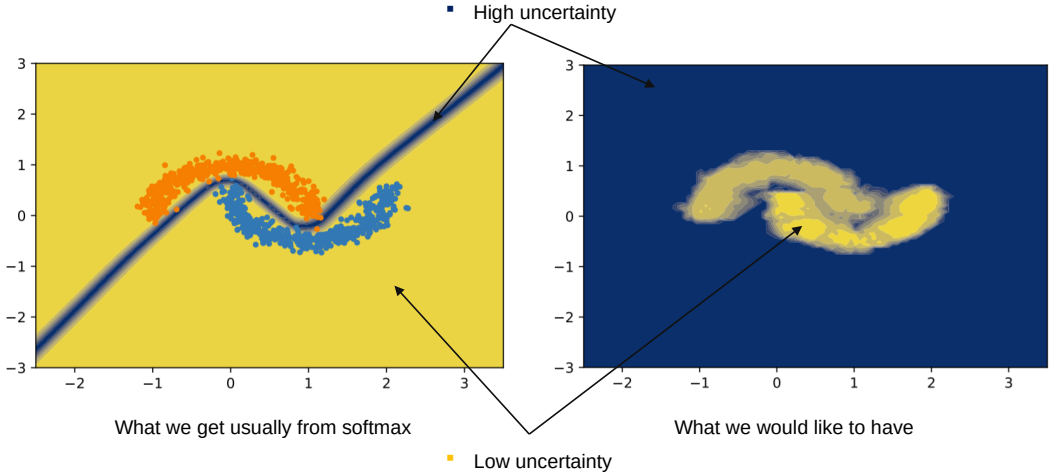
It leads to *Maximum Probability* uncertainty measure:

$$\mathrm{U}_{\mathrm{MP}}(\mathbf{x}) = 1 - \widehat{p}(\mathbf{x}) = 1 - \max_{c} p\left(\mathbf{y} = c \mid \mathbf{x}\right).$$

Why softmax probabilities are bad uncertainty estimates?



Why softmax probabilities are bad uncertainty estimates?



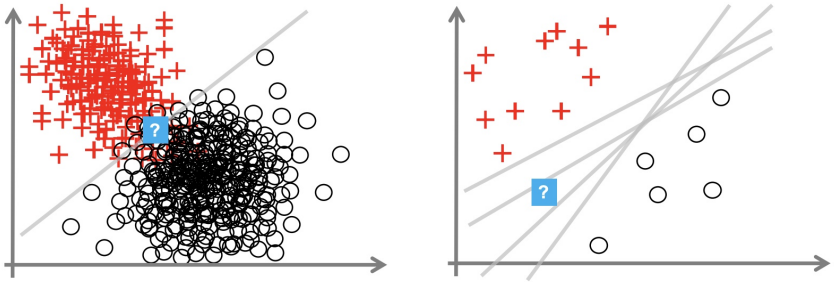
Question: what part of the uncertainty are we missing when using simple probabilities?

Epistemic vs aleatoric uncertainty

The prediction uncertainty can be decomposed into **aleatoric** and **epistemic** terms.

- **Aleatoric** uncertainty reflects noise in data.
  - It could be due to noisy labels, class overlap, or data ambiguity.
  - **This part of the uncertainty can not be reduced when more data is given**.
- **Epistemic** uncertainty reflects lack of knowledge.
  - It is due to the total absence or just a few samples from a particular region.
  - **This part of the uncertainty can be reduced when more data is given**.

Epistemic vs aleatoric uncertainty. Illustration[1]



- **Left:** A case of high aleatoric uncertainty, because the two classes are overlapping.
- **Right:** A case of high epistemic uncertainty due to a lack of knowledge about the right hypothesis.

[1] Eyke Hüllermeier and Willem Waegeman (2021). "Aleatoric and epistemic uncertainty in machine learning: An introduction to concepts and methods". In: *Machine learning*.

Uncertainty quantification for classification

- Consider classification problem over K classes, $\mathbf{y} \in \mathcal{Y} = \{1, \ldots, K\}$.
- Consider vectors of predicted and true label probabilities at point $\mathbf{x}$:

$$\widehat{\eta}_\theta(\mathbf{x}) = p(\mathbf{y} \mid \mathbf{x}, \theta) \quad \text{and} \quad \eta(\mathbf{x}) = p(\mathbf{y} \mid \mathbf{x}).$$

- Introduce a loss function that measures the disagreement between prediction and the true label:

$$\mathcal{L}(\widehat{\eta}_\theta(\mathbf{x}), \mathbf{y}).$$

- It can be, for example:
  - **Zero-one loss**: $\mathcal{L}(\widehat{\eta}_\theta(\mathbf{x}), \mathbf{y}) = \mathbb{1}\left[\mathbf{y} \neq \arg\max_k \left(\widehat{\eta}_\theta(\mathbf{x})\right)_k\right]$;
  - **Cross-Entropy**: $\mathcal{L}(\widehat{\eta}_\theta(\mathbf{x}), \mathbf{y}) = -\left(\log \widehat{\eta}_\theta(\mathbf{x})\right)_{\mathbf{y}}$.

What is uncertainty?

To formalize uncertainties[2], let us first consider a pointwise risk:

$$\mathrm{R}(\widehat{\eta}_\theta \mid \mathbf{x}) = \underbrace{\int \mathcal{L}(\widehat{\eta}_\theta(\mathbf{x}), \mathbf{y})\, dp(\mathbf{y} \mid \mathbf{x})}_{\text{General case}} = \underbrace{\sum_{k=1}^{K} \mathcal{L}(\widehat{\eta}_\theta(\mathbf{x}), k)\, p(\mathbf{y} = k \mid \mathbf{x})}_{\text{Classification}}.$$

Pointwise risk $\mathrm{R}(\widehat{\eta}_\theta \mid \mathbf{x})$ is

- expected error of prediction at point x by model $\widehat{\eta}_\theta$;
- natural uncertainty measure (in expectation);
- not computable in practice directly (as true distribution $p(\mathbf{y} \mid \mathbf{x})$ is not known).

[2] Nikita Kotelevskii et al. (2025). "From risk to uncertainty: Generating predictive uncertainty measures via Bayesian estimation". In: *ICLR 2025*.

What is uncertainty?

Let us consider the following risk decomposition:

$$\underbrace{\mathsf{R}(\widehat{\eta}_\theta \mid \mathbf{x})}_{\text{Total risk}} = \underbrace{\mathsf{R}(\widehat{\eta}_\theta \mid \mathbf{x}) - \mathsf{R}_{\text{Bayes}}(\mathbf{x})}_{\text{Excess risk}} + \underbrace{\mathsf{R}_{\text{Bayes}}(\mathbf{x})}_{\text{Bayes risk}},$$

where $\mathsf{R}_{\text{Bayes}}$ denotes the risk, received by the true data distribution:

$$\mathsf{R}_{\text{Bayes}}(\mathbf{x}) = \int \mathcal{L}(\eta(\mathbf{x}), \mathbf{y})\, dp(\mathbf{y} \mid \mathbf{x}) = \underbrace{\sum_{k=1}^{K} \mathcal{L}(\eta(\mathbf{x}), k)\, p(\mathbf{y} = k \mid \mathbf{x})}_{\text{Classification}}.$$

What is aleatoric uncertainty?

- Bayes risk $\mathrm{R}_{\mathrm{Bayes}}(\mathbf{x}) = \int \mathcal{L}(\eta(\mathbf{x}), \mathbf{y})\, dp(\mathbf{y} \mid \mathbf{x})$:
  - It is the expected error received by the true data generative process.
  - It does not depend on the parameters of the model nor the model architecture.
  - It only depends on the data generative process.
  - Hence, it is associated with **aleatoric uncertainty**.

What is epistemic uncertainty?

- Excess risk $\mathsf{R}_{\mathsf{Exc}}(\widehat{\eta}_\theta \mid \mathbf{x}) = \mathsf{R}(\widehat{\eta}_\theta \mid \mathbf{x}) - \mathsf{R}_{\mathsf{Bayes}}(\mathbf{x})$:
  - It is the difference in risks computed for the approximation and for the true model.
  - It depends on the parameters and the architecture of choice.
  - It expresses how much is approximation worse than the true data generative process.
  - Hence, it can be interpreted as **epistemic uncertainty**.

What is uncertainty?

Consider **Cross-Entropy** loss function: $\mathcal{L}(\widehat{\eta}_\theta(\mathbf{x}), \mathbf{y}) = -\big(\log \widehat{\eta}_\theta(\mathbf{x})\big)_{\mathbf{y}}$.

For this loss function, we have the following equations for risks:

$$\mathsf{R}_{\mathsf{Bayes}}(\mathbf{x}) = \mathbb{H}\big[\eta(\mathbf{x})\big], \qquad \mathsf{R}_{\mathsf{Exc}}(\widehat{\eta}_\theta \mid \mathbf{x}) = \mathsf{KL}\big(\eta(\mathbf{x}) \parallel \widehat{\eta}_\theta(\mathbf{x})\big),$$

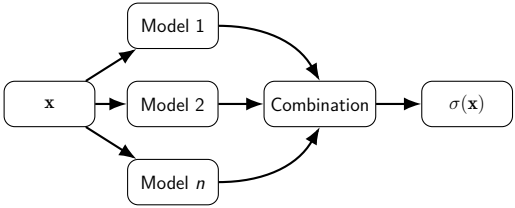
where $\mathbb{H}$ stands for entropy, and KL stands for Kullback-Leibler divergence.

We don't know η in practice. How can we estimate it?

- For **Bayes risk** the answer is easy: just use $\widehat{\eta}_\theta(\mathbf{x})$ as an estimate of $\eta(\mathbf{x})$.
- For **excess risk** we should do something else.

Ensembles

- An ensemble of models is a group of models solving the same task
  - e.g., with same architecture, but different initialization.
- Ensemble prediction averaging is well-known to increase the performance in ML.



- Ensembles[3] can be used not only for prediction but also for uncertainty estimation.
- Further studies[4] showed how to use ensembles for UQ of generative LMs.

[3] Balaji Lakshminarayanan, Alexander Pritzel, and Charles Blundell (2017). "Simple and scalable predictive uncertainty estimation using deep ensembles". In: *NeurIPS 2017*.

[4] Andrey Malinin and Mark J. F. Gales (2021). "Uncertainty Estimation in Autoregressive Structured Prediction". In: *ICLR 2021*.

Uncertainty estimation by disagreement

- Some models provide multiple predictions for a single point:

$$p(\mathbf{y} = k \mid \mathbf{x}, \theta_i), \quad i = 1, \ldots, N.$$

- In this case, we can use disagreement between models as a measure of epistemic uncertainty.

- One possible choice of measure is mutual information:

$$\mathrm{U}_{\mathrm{MI}}(\mathbf{x}) = \frac{1}{N} \sum_{i=1}^{N} \underbrace{\mathbb{H}\left[p(\mathbf{y} \mid \mathbf{x}, \theta_i)\right]}_{\text{Entropy of model } i} - \underbrace{\mathbb{H}\left[\bar{p}\left(\mathbf{y} \mid \mathbf{x}\right)\right]}_{\text{Entropy of averaged model}},$$

where $\bar{p}\left(\mathbf{y} = c \mid \mathbf{x}\right) = \frac{1}{N} \sum_{i=1}^{N} p(\mathbf{y} = c \mid \mathbf{x}, \theta_i)$.

Generative language models

The vast majority of modern language models generate text in the autoregressive way:

$$y_l \sim P(y_l \mid \mathbf{y}_{<l}, \mathbf{x}, \theta), \quad l = 1, \ldots, L,$$

where

- $\mathbf{x}$ is a prompt;
- $\mathbf{y}_{<l} = y_1 \ldots y_{l-1}$;
- θ are parameters of the model.

The resulting probability of the generated sequence $\mathbf{y}$:

$$P(\mathbf{y} \mid \mathbf{x}, \theta) = \prod_{l=1}^{L} P(y_l \mid \mathbf{y}_{<l}, \mathbf{x}, \theta).$$

Generative language models

Probability of the generated sequence $\mathbf{y}$:

$$P(\mathbf{y} \mid \mathbf{x}, \theta) = \prod_{l=1}^{L} P(y_l \mid \mathbf{y}_{<l}, \mathbf{x}, \theta),$$

What is actually y_l or x_t?

$x_t \in \{w_1, \cdots, w_V\}$

What does vocabulary consist of?

Typical size is in 10s of thousands

Character Level

[T] [h] [i] [s] [i] [s] [t] [o] [k] [e] [n] [i] [z] [i] [n] [g] [.]

Word Level

[This] [is] [tokenizing] [.]

Subword Level

[This] [is] [token] [izing] [.]

Peculiarities of UQ for LLMs

- When solving a classification problem for text generated by LLMs, we can quantify the uncertainty using the methods described above.
- However, there are more complex tasks sequence-to-sequence (Seq2Seq) in NLP:
  - machine translation;
  - question answering;
  - abstractive summarization.
- For these problems, we need to compute uncertainty not for a classification label but for the entire generated text.
- Distinguishing epistemic and aleatoric uncertainty is hard as
  - we do not know what our models were trained on (everything is in-domain);
  - ensembles are usually not feasible to train.

Towards well-grounded UQ via Minimum Bayes Risk

Minimum Bayes Risk (MBR) decoding:

$$\mathbf{y}^* = \arg\min_{\mathbf{y}\in\mathcal{Y}} R(\mathbf{y} \mid \mathbf{x}),$$

where $R(\mathbf{y} \mid \mathbf{x})$ is a risk function:

$$R(\mathbf{y} \mid \mathbf{x}) = \mathbb{E}_{\mathbf{y}'\sim p(\mathbf{y}|\mathbf{x})}\ r(\mathbf{y}, \mathbf{y}').$$

with $r(\mathbf{y}, \mathbf{y}')$ being a pairwise loss function.

Natural UQ measure under MBR:

$$\mathrm{U}_{\mathrm{MBR}}(\mathbf{y}^* \mid \mathbf{x}) = \min_{\mathbf{y}\in\mathcal{Y}} R(\mathbf{y} \mid \mathbf{x}) = R(\mathbf{y}^* \mid \mathbf{x}).$$

Towards well-grounded UQ via Minimum Bayes Risk

Consider

$$r(\mathbf{y}, \mathbf{y}') = \mathbb{1}\{\mathbf{y}' \neq \mathbf{y}\}.$$

In this case, the Bayes risk corresponds to the expected zero-one loss of decoding:

$$R_{0/1}(\mathbf{y} \mid \mathbf{x}) = \mathbb{E}_{\mathbf{y}' \sim p(\mathbf{y}|\mathbf{x})} \mathbb{1}\{\mathbf{y}' \neq \mathbf{y}\} = 1 - p(\mathbf{y} \mid \mathbf{x}).$$

This leads to *Sequence Probability (SP)* UQ measure:

$$\mathrm{U}_{\mathrm{SP}}(\mathbf{y}^* \mid \mathbf{x}) = 1 - p\big(\mathbf{y}^* \mid \mathbf{x}\big).$$

Consistency-based uncertainty

- Within MBR framework take $r(\mathbf{y}, \mathbf{y}') = 1 - s(\mathbf{y}, \mathbf{y}')$.
- Now MBR-based uncertainty becomes:

$$\mathrm{U}_{\text{cons}}(\mathbf{y}^* \mid \mathbf{x}) = 1 - \mathbb{E}_{\mathbf{y}' \sim p(\mathbf{y}|\mathbf{x})} s(\mathbf{y}^*, \mathbf{y}').$$

- Consistency-based UQ measure given samples $\{\mathbf{y}^i\}$:

$$\mathrm{U}_{\text{Cons}}(\mathbf{y}^*, \mathbf{x}) = 1 - \frac{1}{N} \sum_{i=1}^{N} s(\mathbf{y}^*, \mathbf{y}^i).$$

Peculiarities of UQ for LLMs. Machine translation



Translate into French language: I want a small cup of coffee



Je veux une petite tasse de café.

Confidence: 100%

Peculiarities of UQ for LLMs. Question answering



Consider there is a human looking for help. How to cure him from pneumonia?



To cure a person suffering from pneumonia, it is important to seek medical attention promptly. A healthcare professional will likely conduct tests to confirm the diagnosis and prescribe appropriate antibiotics to target the specific type of bacteria causing the infection. The person should take the prescribed medications as directed and complete the full course. Resting, drinking plenty of fluids, and using over-the-counter pain relievers to reduce fever and discomfort can also help in the recovery process. Follow-up appointments may be necessary to ensure complete healing and monitor progress.

Confidence: 67%

Different UQ levels for texts

Consider the case of long text generations. Uncertainty Quantification can be:

- **Sequence-level** – too general, does not provide confidence about distinct pieces of information.

Tell me a bio of Tom Hanks.

Tom Hanks is an American actor and filmmaker. He was born on July 9, 1956, in Concord, California. Hanks is considered one of the most talented and versatile actors of his generation and has won numerous awards for his performances in films such as "Forrest Gump", "Philadelphia", and "Saving Private Ryan". He has also been nominated for several Academy Awards. Hanks is also known for his humanitarian work and has been involved in several charitable causes.
Hanks is married to actress Rita Wilson and has four children.
Confidence: 46%

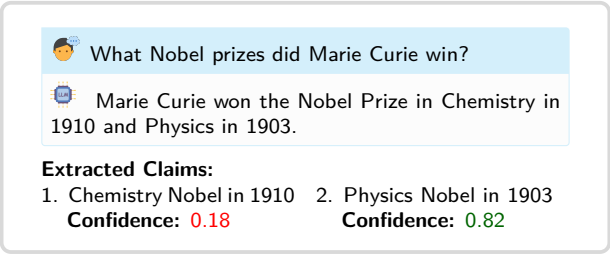
- **Token-level** – too detailed, no factual information.

Tell me a bio of Tom Hanks.

Tom Hanks is an American actor and filmmaker. He was born on July 9, 1956, in Concord, California. Hanks is considered one of the most talented and versatile actors of his generation and has won numerous awards for his performances in films such as "Forrest Gump", "Philadelphia", and "Saving Private Ryan". He has also been nominated for several Academy Awards. Hanks is also known for his humanitarian work and has been involved in several charitable causes.
Hanks is married to actress Rita Wilson and has four children.

Claim-level uncertainty quantification

Claim-level is more representative:



Numerical confidence for each extracted claim.

Claim-level UQ pipeline:

- Extract claims from the generated text
- For each claim, get the tokens in the generation related to this claim
- Perform Uncertainty Quantification for each token
- Aggregate uncertainties of tokens into claim uncertainty

Types of uncertainty quantification methods

- Unsupervised methods
  - Typically are based on some intuitive statistic that reflects predictive uncertainty.
  - Example: sequence probability $P(\mathbf{y} \mid \mathbf{x})$.
- Supervised methods
  - Fit function that predicts erroneous generations based on labeled dataset.
  - Example: linear regression over embedding vectors of the model.

Types of unsupervised UQ methods

Black-box methods

- Verbalized uncertainty
  - Directly asking the model about its confidence in a generated answer

*There exist white-box variations of verbalized uncertainty methods.

Types of unsupervised UQ methods

Black-box methods

- Verbalized uncertainty
  - Directly asking the model about its confidence in a generated answer
- Consistency-based
  - Sample multiple generations and measure their (semantic) consistency

*There exist white-box variations of verbalized uncertainty methods.

Types of unsupervised UQ methods

Black-box methods

- Verbalized uncertainty
  - Directly asking the model about its confidence in a generated answer
- Consistency-based
  - Sample multiple generations and measure their (semantic) consistency

White-box methods

- Information-theoretic
  - Assess uncertainty as measured by probabilities given by the model

*There exist white-box variations of verbalized uncertainty methods.

Types of unsupervised UQ methods

Black-box methods

- Verbalized uncertainty
  - Directly asking the model about its confidence in a generated answer
- Consistency-based
  - Sample multiple generations and measure their (semantic) consistency

White-box methods

- Information-theoretic
  - Assess uncertainty as measured by probabilities given by the model
- Introspective
  - Analyze model embeddings and/or attention masks

*There exist white-box variations of verbalized uncertainty methods.

Types of unsupervised UQ methods

Black-box methods

- Verbalized uncertainty
  - Directly asking the model about its confidence in a generated answer
- Consistency-based
  - Sample multiple generations and measure their (semantic) consistency

White-box methods

- Information-theoretic
  - Assess uncertainty as measured by probabilities given by the model
- Introspective
  - Analyze model embeddings and/or attention masks

Hybrid methods

- Combinations of various methods above

*There exist white-box variations of verbalized uncertainty methods.

LM-Polygraph: Whitebox unsupervised UQ methods

Type	Uncertainty Quantification Method	Category	Compute	Needs Training Data	Level
White-box	Maximum Sequence Probability (MSP)	Information-theoretic	Low	No	Seq./claim
	Perplexity (Fomicheva et al. 2020)		Low	No	Seq./claim
	Mean/Max Token Entropy (TE; Fomicheva et al. 2020)		Low	No	Seq./claim
	TokenSAR (Duan et al. 2024)		Low	No	Seq.
	FrequencyScoring (Mohri and Hashimoto 2024)		High	No	Seq./claim
	EigenScore (Chen et al. 2024)		High	No	Seq./claim
	CCP (Fadeeva et al. 2024)		Low	No	Seq./claim
	Monte Carlo Sequence Entropy (MC-SE; Kuhn, Gal, and Farquhar 2023)		High	No	Seq.
	Monte Carlo Norm. Seq. Entropy (MC-NSE; Malinin and Gales 2021)		High	No	Seq.
	Semantic Entropy (Kuhn, Gal, and Farquhar 2023)		High	No	Seq.
	SentenceSAR (Duan et al. 2024)		High	No	Seq.
	SAR (Duan et al. 2024)		High	No	Seq.
	CoCoA (Vashurin et al. 2025c)	Hybrid	High	No	Seq.
	Mahalanobis Distance (MD; Lee et al. 2018)	Introspective	Low	Yes	Seq.
	Robust Density Estimation (RDE; Yoo et al. 2022)		Low	Yes	Seq.
	Relative Mahalanobis Distance (RMD; Ren et al. 2023)		Low	Yes	Seq.
	Hybrid Uncertainty Quantification (HUQ; Vazhentsev et al. 2023)		Low	Yes	Seq.
	Attention Score (Sriramanan et al. 2024a)		Medium	No	Seq./claim
	RAUQ (Vazhentsev et al. 2025b)		Low	No	Seq./claim

LM-Polygraph: Blackbox unsupervised UQ methods

Type	Uncertainty Quantification Method	Category	Compute	Needs Training Data	Level
Black-box	Number of Semantic Sets (NumSet; Lin, Trivedi, and Sun 2024)	Consistency-based	High	No	Seq.
	Sum of Eigenvalues of Laplacian (EigV; Lin, Trivedi, and Sun 2024)		High	No	Seq.
	Degree Matrix (Deg; Lin, Trivedi, and Sun 2024)		High	No	Seq.
	Eccentricity (Ecc; Lin, Trivedi, and Sun 2024)		High	No	Seq.
	Lexical Similarity (LexSim; Fomicheva et al. 2020)		High	No	Seq.
	LUQ (Zhang et al. 2024)		High	No	Seq.
	Kernel Language Entropy (Nikitin et al. 2024)		High	No	Seq.
	Semantic Density (Qiu and Miikkulainen 2024)		High	No	Seq.
	BB Semantic Entropy (Farquhar et al. 2024)		High	No	Seq.
	Verbalized 1S (Tian et al. 2023)	Verbalized	Low	No	Seq.
	Verbalized 2S (Tian et al. 2023)		Medium	No	Seq.

Takeaways & QA

- Predictive uncertainty of the model has different sources (aleatoric vs epistemic).
- Uncertainty quantification for language models has peculiarities compared to classification/regression.
- UQ methods can be constructed in supervised or unsupervised way.
- Multiple types of UQ methods exist which are based on various levels of information from the model.

3. Information-Theoretic, Consistency-based, and Verbalized Uncertainty

Roman Vashurin

Verbalized uncertainty

- The most straightforward way to estimate the uncertainty in LLMs is to **ask them directly**.
- Requests the numeric uncertainty to be directly reported by the model.
- Relies on the ability of the LLM to assess its own uncertainty – low granularity, scale-dependent.

Verbalized uncertainty[5]

- Example of a prompt:

```
Provide your best guess and the probability that it is correct
(0.0 to 1.0) for the following question. For example:

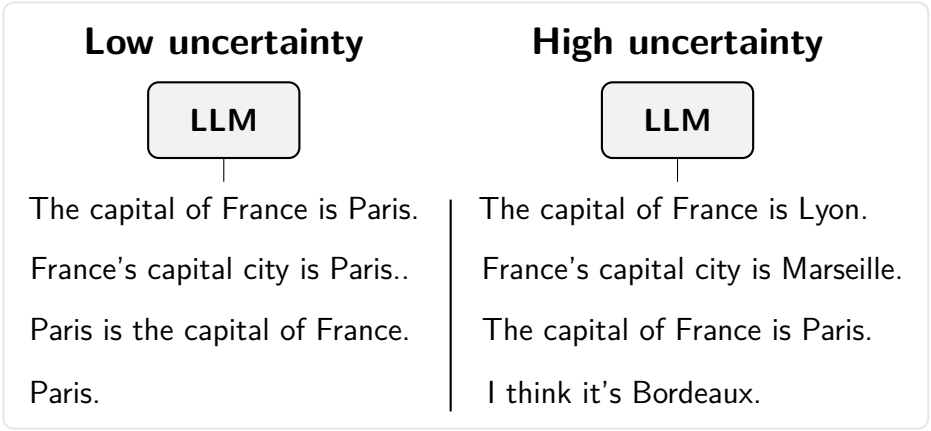
Guess: <most likely guess>
Probability: <the probability between 0.0 and 1.0 that
your guess is correct>

Question: Who was the first president of the United States?
```

[5] Katherine Tian et al. (2023). "Just Ask for Calibration: Strategies for Eliciting Calibrated Confidence Scores from Language Models Fine-Tuned with Human Feedback". In: *EMNLP 2023*.

Consistency-based uncertainty

- **Intuition:** diverse responses to the same prompt indicate high uncertainty.



Frequency scoring

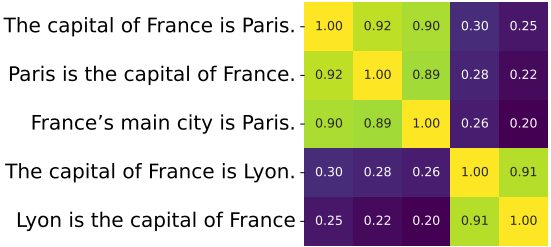
FrequencyScoring[6] estimates the uncertainty of a generation $\mathbf{y}^*$ by analyzing its support in other generation $\mathbf{y}^j$:

$$\mathrm{U}_{\mathrm{FreqScore}}(\mathbf{y}^*, \mathbf{x}) = \sum_{j=1}^{N} \mathbb{1}\left[\mathsf{NLI}(\mathbf{y}^*, \mathbf{y}^j) = \text{`c'}\right] - \sum_{j=1}^{N} \mathbb{1}\left[\mathsf{NLI}(\mathbf{y}^*, \mathbf{y}^j) = \text{`e'}\right].$$

[6] Christopher Mohri and Tatsunori Hashimoto (2024). "Language Models with Conformal Factuality Guarantees". In: *ICML 2024*, Originally proposed for claim-level, with GPT-4 serving as NLI model.

Similarity-based uncertainty

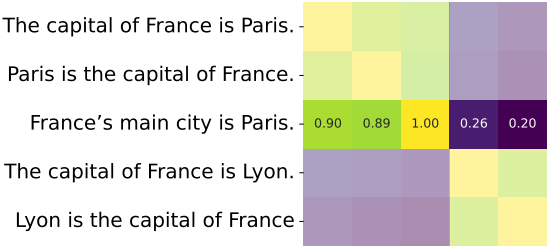
- Consider a set of sampled outputs: $\mathbf{y}^i \sim p(\mathbf{y} \mid \mathbf{x}), \quad i = 1, \ldots, N.$
- Similarity matrix: $S_{ij} = s(\mathbf{y}^i, \mathbf{y}^j)$, where s can be ROUGE-L, BLEU, NLI, ...



Similarity-based uncertainty

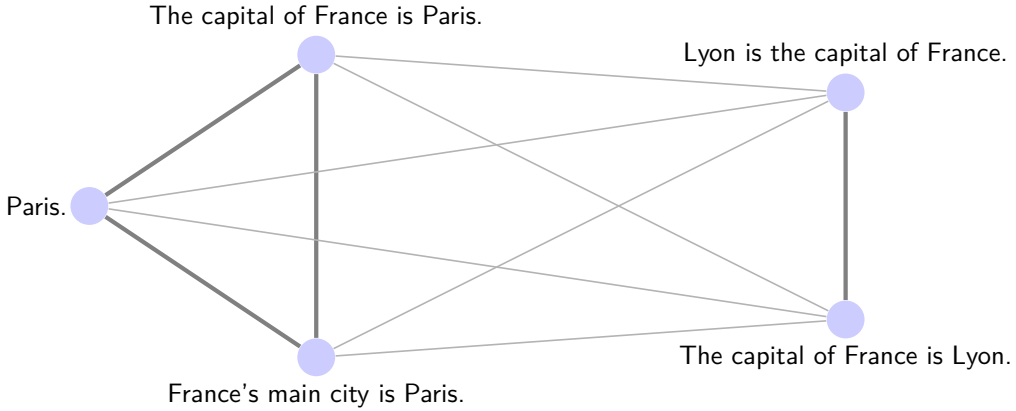
- Within MBR framework take $r(\mathbf{y}, \mathbf{y}') = 1 - s(\mathbf{y}, \mathbf{y}')$.
- Consistency-based UQ measure given samples $\{\mathbf{y}^i\}$:

$$\mathrm{U}_{\mathrm{Cons}}(\mathbf{y}^*, \mathbf{x}) = 1 - \frac{1}{N} \sum_{i=1}^{N} s(\mathbf{y}^*, \mathbf{y}^i).$$



Similarity as a graph

- Sampled sequences are nodes, pairwise similarities are edges.
- Then similarity matrix S becomes an adjacency matrix of the graph.



Graph-based uncertainty

- Normalized Graph Laplacian: $L = I - D^{-1/2} S D^{-1/2}$, where D is the corresponding degree matrix of the graph, and I is the identity matrix.
- Eigenvalues of graph Laplacian represent a measure of graph connectivity[7]:

$$\mathrm{U}_{\mathrm{EigV}}(\mathbf{x}) = \sum_{k=1}^{N} \max\big(0, 1 - \lambda_k(\mathbf{x})\big),$$

where $\lambda_k(\mathbf{x})$ is the k-th eigenvalue of L.

[7] Zhen Lin, Shubhendu Trivedi, and Jimeng Sun (2024). "Generating with Confidence: Uncertainty Quantification for Black-box Large Language Models". In: *TMLR 2024*.

Information-theoretic uncertainty

- **Sequence Probability** (log-probability) is the most straightforward measure of uncertainty for a response $\mathbf{y}^*$:

$$\mathrm{U}_{\mathrm{SP}}(\mathbf{y}^*, \mathbf{x}) = -\log P\big(\mathbf{y}^* \mid \mathbf{x}\big) = -\sum_{j=1}^{L} \log P\big(y_j^* \mid \mathbf{y}_{<j}^*, \mathbf{x}\big).$$

[8] Roman Vashurin et al. (2025b). "UNCERTAINTY-LINE: Length-Invariant Estimation of Uncertainty for Large Language Models". In: *EMNLP 2025*.

Information-theoretic uncertainty

- **Sequence Probability** (log-probability) is the most straightforward measure of uncertainty for a response $\mathbf{y}^*$:

$$\mathrm{U}_{\mathrm{SP}}(\mathbf{y}^*, \mathbf{x}) = -\log P\big(\mathbf{y}^* \mid \mathbf{x}\big) = -\sum_{j=1}^{L} \log P\big(y_j^* \mid \mathbf{y}_{<j}^*, \mathbf{x}\big).$$

- Due to autoregressive probabilistic model, it has a **natural bias towards shorter sequences**.

[8] Roman Vashurin et al. (2025b). "UNCERTAINTY-LINE: Length-Invariant Estimation of Uncertainty for Large Language Models". In: *EMNLP 2025*.

Information-theoretic uncertainty

- **Sequence Probability** (log-probability) is the most straightforward measure of uncertainty for a response $\mathbf{y}^*$:

$$\mathrm{U}_{\mathrm{SP}}(\mathbf{y}^*, \mathbf{x}) = -\log P\big(\mathbf{y}^* \mid \mathbf{x}\big) = -\sum_{j=1}^{L} \log P\big(y_j^* \mid \mathbf{y}_{<j}^*, \mathbf{x}\big).$$

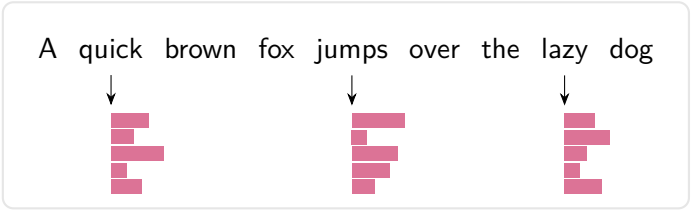
- Due to autoregressive probabilistic model, it has a **natural bias towards shorter sequences**.
- **Perplexity** (length-normalized log-probability) partially mitigates this bias[8]:

$$\mathrm{U}_{\mathrm{PPL}}(\mathbf{y}^*, \mathbf{x}) = -\log \bar{P}(\mathbf{y}^* \mid \mathbf{x}) = -\frac{1}{L}\sum_{j=1}^{L} \log P\big(y_j^* \mid \mathbf{y}_{<j}^*, \mathbf{x}\big).$$

[8] Roman Vashurin et al. (2025b). "UNCERTAINTY-LINE: Length-Invariant Estimation of Uncertainty for Large Language Models". In: *EMNLP 2025*.

Mean token entropy

- In the white-box setting, full distributions over vocabulary are available at each token position:



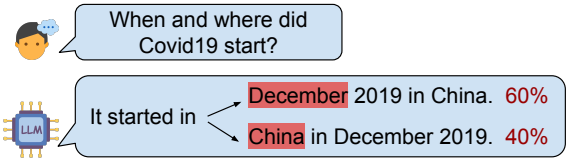
- At each position we can compute entropy of these distributions, and their average gives **Mean Token Entropy** (MTE):

$$\mathrm{U}_{\mathrm{MTE}}(\mathbf{y}^*, \mathbf{x}) = -\frac{1}{L}\sum_{j=1}^{L}\sum_{t\in\mathcal{V}} P(y_j^* = t \mid \mathbf{y}^*_{<j}, \mathbf{x}) \log P(y_j^* = t \mid \mathbf{y}^*_{<j}, \mathbf{x}).$$

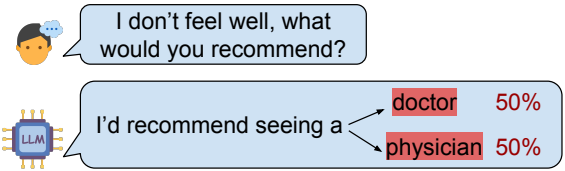
Variability is not always relevant for hallucination detection

Problem: probability-based methods do not account for different types of uncertainties. $P(y_j \mid \mathbf{y}_{<j}, \mathbf{x})$ can be low because model is not confident of:

which claim to mention at current token — *Claim type/order uncertainty*:



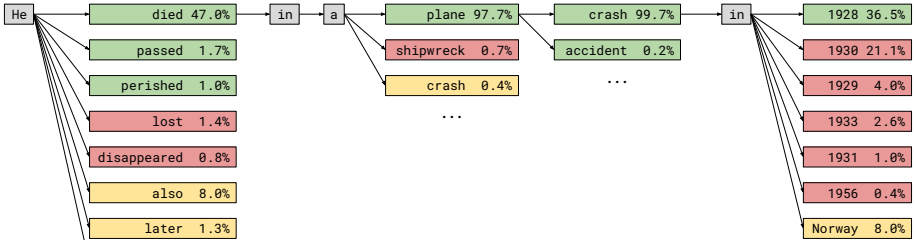
which words to use to mention the claim — *Surface form uncertainty*:



Variability is not always relevant for hallucination detection

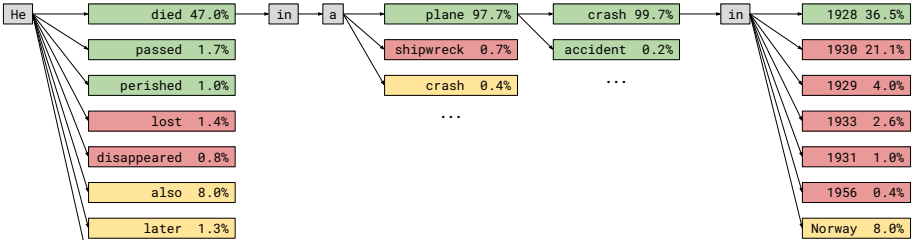
Example part of model generation. Each non-functional token is supplied with top alternative tokens in the generation.

- Synonyms/hypernyms reflect *surface form uncertainty*
- Neutral words reflect *claim type/order uncertainty*.



Claim-Conditioned Probability: CCP

Solution: Claim-Conditioned Probability[9]



$$\mathrm{CCP}(y_j^*, \mathbf{y}_{<j}^*, \mathbf{x}) = \frac{\sum P(\quad)}{\sum P(\quad + \quad)}$$

[9] Ekaterina Fadeeva et al. (2024). "Fact-Checking the Output of Large Language Models via Token-Level Uncertainty Quantification". In: *Findings of ACL 2024*.

Claim-Conditioned Probability: CCP

Solution: Claim-Conditioned Probability[10]

$$\mathrm{CCP}(y_j^*, \mathbf{y}_{<j}^*, \mathbf{x}) = \frac{\sum_{t \in \mathcal{V}} \mathbb{1}\left[\texttt{NLI}(t, y_j^*) = \texttt{`e'}\right] \; P(y_j^* = t \mid \mathbf{y}_{<j}^*, \mathbf{x})}{\sum_{t \in \mathcal{V}} \mathbb{1}\left[\texttt{NLI}(t, y_j^*) \in \{\texttt{`e'}, \texttt{`c'}\}\right] \; P(y_j^* = t \mid \mathbf{y}_{<j}^*, \mathbf{x})}.$$

Token-level CCP scores are aggregated to claim- or sequence-level uncertainty:

$$\mathrm{U}_{\mathrm{CCP}}(\mathbf{y}^*, \mathbf{x}) = -\sum_{j=1}^{L} \log \mathrm{CCP}(y_j^*, \mathbf{y}_{<j}^*, \mathbf{x}).$$

[10] Ekaterina Fadeeva et al. (2024). "Fact-Checking the Output of Large Language Models via Token-Level Uncertainty Quantification". In: *Findings of ACL 2024*.

Not all tokens contribute to uncertainty equally

- **Problem:** Natural language contains lots of service parts, and modern LLMs tend to augment answers with preambles:

  "The answer is: Donald Trump is the current president of the United States."

  While these do not contribute much to the meaning of the output, the measures we've discussed so far treat all tokens in the output equally.

Not all tokens contribute to uncertainty equally

- **Problem:** Natural language contains lots of service parts, and modern LLMs tend to augment answers with preambles:

  "The answer is: Donald Trump is the current president of the United States."

  While these do not contribute much to the meaning of the output, the measures we've discussed so far treat all tokens in the output equally.

- **Solution:** Put weights on the tokens according to their contribution to the overall meaning of the output:

  The answer is: Donald Trump is the current president of the United States.

Info-theoretic confidence with token emphasis

- Weighted-token uncertainty:

$$\mathrm{U}_{\mathrm{Seq}}(\mathbf{y}^*, \mathbf{x}) = \sum_{j=1}^{L} w_j \mathrm{U}_{\mathrm{tok}}(y_j \mid y_{<j}, \mathbf{x}),$$

  where w_j is the weight associated with the j-th token in the sequence $\mathbf{y}^*$.

- Weights can be determined by:
  - NLI (TokenSAR[11])
  - Named entity extraction (Focus[12])

[11] Jinhao Duan et al. (2024). "Shifting Attention to Relevance: Towards the Predictive Uncertainty Quantification of Free-Form Large Language Models". In: *ACL 2024*.

[12] Tianhang Zhang et al. (2023). "Enhancing Uncertainty-Based Hallucination Detection with Stronger Focus". In: *EMNLP 2023*.

Monte-Carlo sequence entropy (MCSE/MCNSE)

- **Monte Carlo approximation** of sequence entropy[13] with N samples $\mathbf{y}^i \sim P(\mathbf{y} \mid \mathbf{x})$:

$$\mathrm{U}_{\mathrm{MCSE}}(\mathbf{x}) = -\frac{1}{N} \sum_{i=1}^{N} \log P\big(\mathbf{y}^i \mid \mathbf{x}\big).$$

- As with sequence probability, MCSE has inherent bias toward shorter sequences due to autoregressive nature.

[13] Andrey Malinin and Mark J. F. Gales (2021). "Uncertainty Estimation in Autoregressive Structured Prediction". In: *ICLR 2021*.

Monte-Carlo sequence entropy (MCSE/MCNSE)

- **Monte Carlo approximation** of sequence entropy[13] with N samples $\mathbf{y}^i \sim P(\mathbf{y} \mid \mathbf{x})$:

$$\mathrm{U}_{\mathrm{MCSE}}(\mathbf{x}) = -\frac{1}{N}\sum_{i=1}^{N} \log P\left(\mathbf{y}^i \mid \mathbf{x}\right).$$

- As with sequence probability, MCSE has inherent bias toward shorter sequences due to autoregressive nature.
- In the same fashion, to mitigate it, **length-normalized log-probability** $\bar{P}\left(\mathbf{y}^i \mid \mathbf{x}\right)$ can be used:

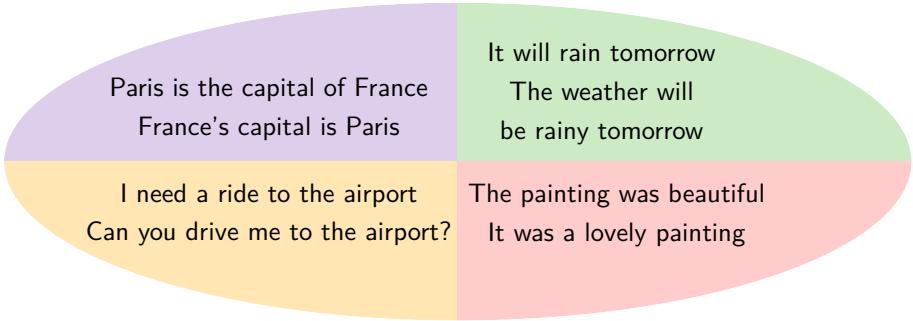
$$\mathrm{U}_{\mathrm{MCNSE}}(\mathbf{x}) = -\frac{1}{N}\sum_{i=1}^{N} \log \bar{P}\left(\mathbf{y}^i \mid \mathbf{x}\right).$$

[13] Andrey Malinin and Mark J. F. Gales (2021). "Uncertainty Estimation in Autoregressive Structured Prediction". In: *ICLR 2021*.

Adding semantics

- **Key idea:** different sequences can have the same meaning.
- Consider a semantic set:

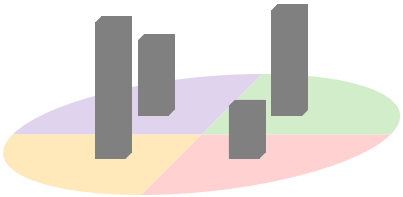
$$\mathcal{C} = \{\mathbf{y} \colon \forall \mathbf{y}' \in \mathcal{C}, \mathrm{NLI}(\mathbf{y}, \mathbf{y}') = \mathrm{NLI}(\mathbf{y}', \mathbf{y}) = \textit{entail}\}.$$



Semantic entropy

- Entropy over semantic clusters[14].
- Let $\{\mathcal{C}_m\}_{m=1}^{M}$ be semantic clusters, then

$$\mathrm{U_{SE}} = -\frac{1}{N}\sum_{m=1}^{M} |\mathcal{C}_m| \log \hat{P}_m(\mathbf{x}), \quad \hat{P}_m(\mathbf{x}) = \sum_{\mathbf{y}\in\mathcal{C}_m} P(\mathbf{y} \mid \mathbf{x}).$$

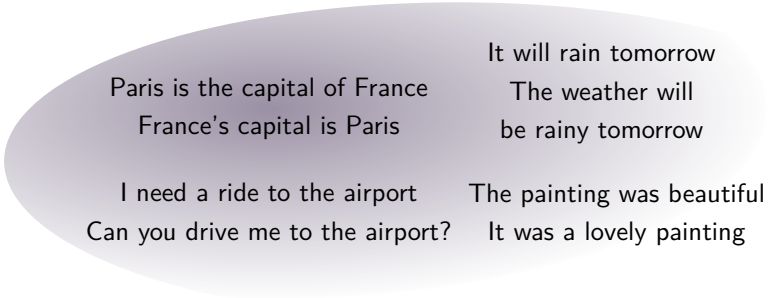


[14] Lorenz Kuhn, Yarin Gal, and Sebastian Farquhar (2023). "Semantic Uncertainty: Linguistic Invariances for Uncertainty Estimation in Natural Language Generation". In: *ICLR 2023*.

Shifting Attention to Relevance: SAR

- Soft clustering through relevance term R_S, relaxing hard partition assumption:

$$R_S(\mathbf{y}^i, \mathbf{x}) = \sum_{j \neq i} s(\mathbf{y}^i, \mathbf{y}^j) P(\mathbf{y}^j \mid \mathbf{x}).$$



Shifting Attention to Relevance: SAR

- Soft clustering through relevance term R_S, relaxing hard partition assumption:

$$R_S(\mathbf{y}^i, \mathbf{x}) = \sum_{j \neq i} s(\mathbf{y}^i, \mathbf{y}^j) P(\mathbf{y}^j \mid \mathbf{x}).$$

- Generalization of semantic entropy with soft clustering through R_S[15]:

$$\mathrm{U}_{\mathrm{SentSAR}}(\mathbf{x}) = -\frac{1}{N} \sum_{i=1}^{N} \log\big(P(\mathbf{y}^i \mid \mathbf{x}) + \tfrac{1}{t} R_S(\mathbf{y}^i, \mathbf{x})\big).$$

[15] Jinhao Duan et al. (2024). "Shifting Attention to Relevance: Towards the Predictive Uncertainty Quantification of Free-Form Large Language Models". In: *ACL 2024*.

CoCoA

- **Key idea:** combine information theoretic and consistency-based methods via MBR:

$$r(\mathbf{y}, \mathbf{y}' \mid \mathbf{x}) = \mathrm{U}_{\mathrm{inf}}(\mathbf{y}, \mathbf{x}) \cdot \big(1 - s(\mathbf{y}, \mathbf{y}')\big),$$

- $\mathrm{U}_{\mathrm{inf}}$ can be any information-theoretic uncertainty estimate.
- The resulting CoCoA[16] UQ measure becomes:

$$\mathrm{U}_{\mathrm{CoCoA}}(\mathbf{y}^*, \mathbf{x}) = \mathrm{U}_{\mathrm{inf}}(\mathbf{y}^*, \mathbf{x}) \cdot \mathrm{U}_{\mathrm{cons}}(\mathbf{y}^*, \mathbf{x}).$$

- For example, one can consider:

$$\mathrm{U}_{\mathrm{CoCoA}}(\mathbf{y}^*, \mathbf{x}) = \Big[-\log P(\mathbf{y}^* \mid \mathbf{x})\Big] \cdot \frac{1}{N} \sum\nolimits_{i=1}^{N} \big(1 - s(\mathbf{y}^*, \mathbf{y}^i)\big).$$

[16] Roman Vashurin et al. (2025c). "Uncertainty Quantification for LLMs through Minimum Bayes Risk: Bridging Confidence and Consistency". In: *NeurIPS 2025*.

Comparative Performance

UQ Method	Llama-3.1 8B			Gemma-3 12B			Mean
	QA	ATS	NMT	QA	ATS	NMT	
Sequence Probability	.353	.173	.395	.382	.317	.479	.350
Perplexity	.392	**.354**	.402	.456	.334	.548	.414
Mean Token Entropy	.335	.336	.437	.447	.311	.576	.407
MC SE	.330	.118	.382	.394	.060	.346	.272
MC NSE	.343	.180	.384	.369	.085	.428	.298
NumSemSets	.328	.136	.047	.356	.114	.038	.170
Semantic Entropy	.339	.119	.393	.392	.048	.353	.274
EigVal	.369	.133	.391	.399	.084	.295	.279
Semantic Density	.136	.316	.382	.139	.353	.405	.289
CoCoA	**.413**	.283	**.506**	**.464**	**.385**	**.626**	**.446**

Table: Mean PRR↑ across tasks for the evaluated LLMs

- CoCoA shows the strongest overall performance, consistently ranking one of the best by average performance in a task.

Sampling considerations

- Sequence entropy estimators assume **unbiased stochastic sampling** from $P(\mathbf{y} \mid \mathbf{x})$.

- Deterministic decoding requires correction specific to the particular decoding bias. For beam search, given obtained set of sequences $\mathcal{B}$, becomes[17]:

$$\mathbf{y}^i \in \mathcal{B}, \qquad \pi_i = \frac{P\left(\mathbf{y}^i \mid \mathbf{x}\right)}{\sum_{j=1}^{N} P\left(\mathbf{y}^j \mid \mathbf{x}\right)}.$$

[17] Andrey Malinin and Mark J. F. Gales (2021). "Uncertainty Estimation in Autoregressive Structured Prediction". In: *ICLR 2021*.

Beam search–based consistency estimation[18]

- Beam Search has other surprising advantages for UQ!
- Use corrected probabilities in the following estimator:

$$\mathrm{U}_{\mathrm{cons}}^{\mathrm{beam}}(\mathbf{y}^*, \mathbf{x}) = 1 - \sum_{i=1}^{N} \pi_i \, s(\mathbf{y}^*, \mathbf{b}^i)$$

- Same construction applies to other consistency-based methods, e.g. graph-based and CoCoA

[18] Ekaterina Fadeeva et al. (2025). "Don't Throw Away Your Beams: Improving Consistency-based Uncertainties in LLMs via Beam Search". In: *arXiv preprint arXiv:2512.09538*.

When beam search beats multinomial sampling[19]

- Let the total probability mass covered by the beam set be

$$m_{\mathcal{B}} = \sum_{i=1}^{N} P(\mathbf{b}^i \mid \mathbf{x})$$

- Distribution-free result: the beam-weighted estimator $\mathrm{U}_{\mathrm{cons}}^{\mathrm{beam}}$ has lower error than multinomial sampling $\mathrm{U}_{\mathrm{cons}}$ whenever

$$m_{\mathcal{B}} > 1 - \frac{1}{2\sqrt{N}}$$

- For $N = 10$, this requires only $m_{\mathcal{B}} > 0.84$

[19] Ekaterina Fadeeva et al. (2025). "Don't Throw Away Your Beams: Improving Consistency-based Uncertainties in LLMs via Beam Search". In: *arXiv preprint arXiv:2512.09538*.

How much better is Beam Search?

UQ Method	Llama-3.1 8B short-form QA	Qwen-3 8B short-form QA	Mean
Sequence Probability	.344	.289	.316
Perplexity	.323	.276	.300
Mean Token Entropy	.364	.297	.330
Dissimilarity	.379	.327	.353
Dissimilarity + beamsearch	.417 ↑	**.355** ↑	**.386** ↑
Eccentricity	.368	.251	.310
Eccentricity + beamsearch	.397 ↑	.345 ↑	.371 ↑
Cocoa	.404	.334	.369
Cocoa + beamsearch	**.426** ↑	.347 ↑	**.386** ↑

Table: Mean PRR↑ across tasks for the evaluated LLMs

- Beam search versions of consistency-based UQ improve performance over their multinomial sampling counterparts for short-form QA.

Code example in LM-Polygraph

```
from transformers import AutoModel, AutoTokenizer
from lm_polygraph import estimate_uncertainty
from lm_polygraph.model_adapters import WhiteboxModel
from lm_polygraph.estimators import *

model_path = "meta-llama/Llama-3.1-8B-Instruct"
base_model = AutoModel.from_pretrained(model_path)
tokenizer = AutoTokenizer.from_pretrained(model_path)

model = WhiteboxModel(model_path, tokenizer)

ue_method = CocoaMSP()
input_text = "Who is George Bush?"
estimate_uncertainty(model, ue_method, input_text)
```

UQ method is specified here

Takeaways & QA

- Uncertainty estimation in unsupervised setting can be tackled from various directions:
  - Elicitation of verbalized uncertainty
  - Consistency analysis
  - Information-theoretic analysis
  - Hybrid measures
- Generally, hybrid measures perform better, as info-theoretic and consistency-based approaches complement each other nicely.
- CoCoA shows that a very simple multiplicative combination of the two types of measures can be quite efficient.
- In certain cases beam search outperforms sampling.

Break 30 minutes

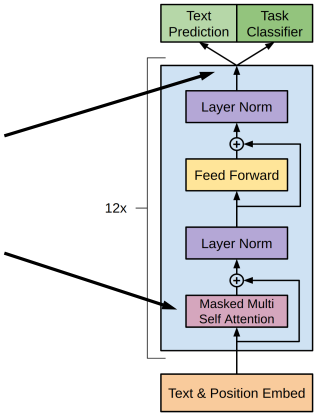
4. Introspective Methods

Artem Vazhentsev

Inside LLM

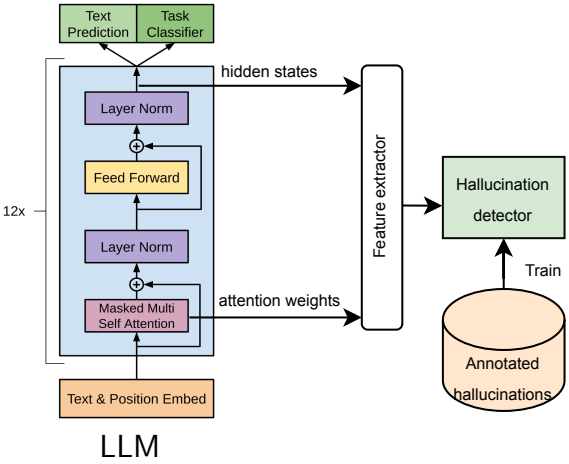
Uncertainty quantification methods can leverage two **internal signals** from LLMs:

- Hidden states. The embedding vectors from each decoder layer for every generated token.
- Attention weights. The lower-triangular matrices, which illustrate how each token attends to the previous tokens during generation.



Transformer Decoder Architecture

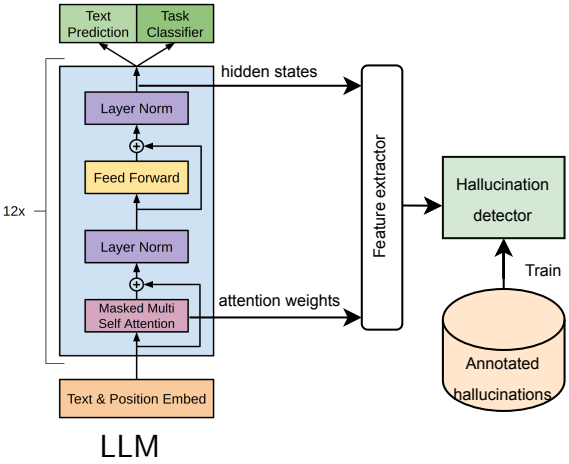
From manual engineering to data-driven methods



Intuition: Let's bypass all UQ challenges by learning from the data!

1. Create features from internal states of the LLM.
2. Collect training data: hallucinations / non-hallucinations.
3. Train a supervised hallucination detector.

From manual engineering to data-driven methods

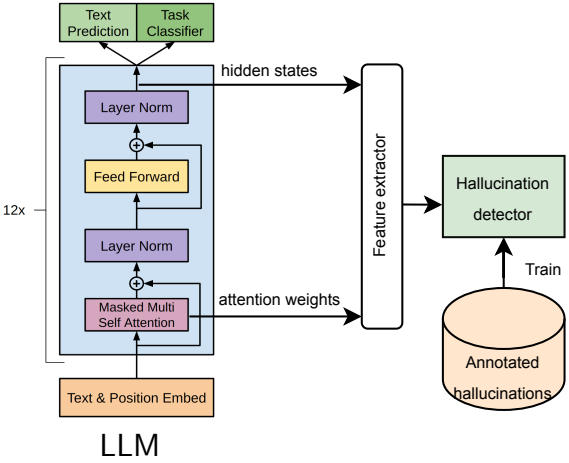


Intuition: Let's bypass all UQ challenges by learning from the data!

1. Create features from internal states of the LLM.
2. Collect training data: hallucinations / non-hallucinations.
3. Train a supervised hallucination detector.

Question: How is it different with an external fact-checking tool?

From manual engineering to data-driven methods



Intuition: Let's bypass all UQ challenges by learning from the data!

1. Create features from internal states of the LLM.
2. Collect training data: hallucinations / non-hallucinations.
3. Train a supervised hallucination detector.

Question: How is it different with an external fact-checking tool?

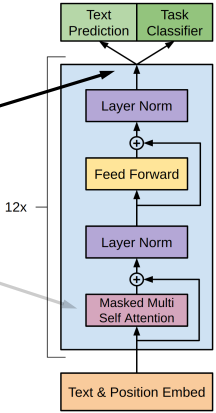
Answer: External fact-checker uses LLM outputs, supervised uncertainty quantification uses LLM internal states.

4.1. Methods Based on Hidden States

Inside LLM: Hidden states

Uncertainty quantification methods can leverage two **internal signals** from LLMs:

- Hidden states. The embedding vectors from each decoder layer for every generated token.
- Attention weights. The lower-triangular matrices, which illustrate how each token attends to the previous tokens during generation.

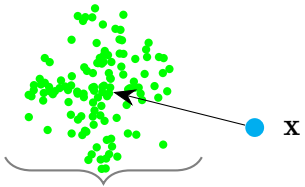


Transformer Decoder Architecture

Inside LLM: Hidden states

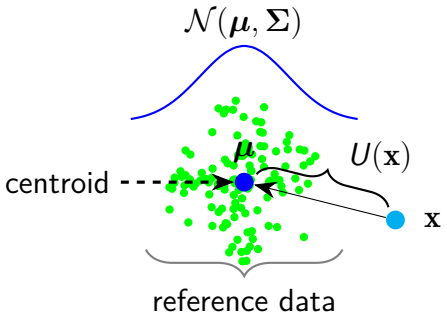
Intuition:

- Softmax always gives probabilities that sum to one.
- Unusual behaviour might be easier to observe before the softmax, i.e. looking at embeddings (hidden states) drawn from intermediate LLM layers.
- Hidden states define a feature space in which prompts that lead to hallucinations and factual answers are readily separable.
- One can measure the distance from the prompt $\mathbf{x}$ to a "reference" distribution.



Density-based methods

Idea: fit an auxiliary feature-space distribution using the hidden states of the model.



Mahalanobis distance

Mahalanobis distance (MD)[20] is proportional to the negative log-likelihood of a multivariate normal distribution, up to an additive constant:

$$\mathrm{U}_{\mathrm{MD}}(\mathbf{x}, l) = (h_l(\mathbf{x}) - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (h_l(\mathbf{x}) - \boldsymbol{\mu}).$$

Tested instance

Hidden representation of $\mathbf{x}$ from layer l

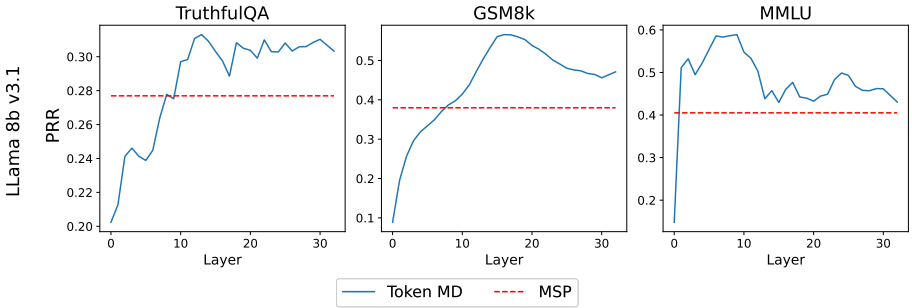
Covariance matrix

Centroid of the reference data

[20] Kimin Lee et al. (2018). "A simple unified framework for detecting out-of-distribution samples and adversarial attacks". In: *NeurIPS 2018*.

Layer-wise Mahalanobis distance

- MD that uses embeddings from intermediate layers yields better results than the MSP baseline[21].



[21] Artem Vazhentsev et al. (2025a). "Token-Level Density-Based Uncertainty Quantification Methods for Eliciting Truthfulness of Large Language Models". In: *NAACL 2025*.

Modifications of MD

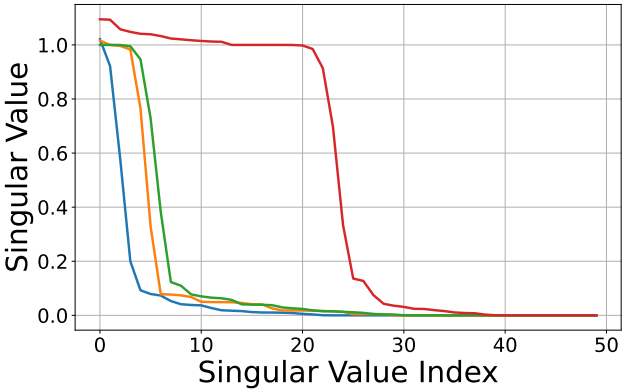
- Remove outliers from the training dataset used for parameter estimation in MD by applying the Minimum Covariance Determinant (MCD) estimator[22].
- Incorporate a background contrastive MD score $U^0_{\mathrm{MD}}(\mathbf{x}, l)$, which measures the distance from a broad distribution approximating all domains[23].

[22] KiYoon Yoo et al. (2022). "Detection of Adversarial Examples in Text Classification: Benchmark and Baseline via Robust Density Estimation". In: *Findings of ACL 2022*.

[23] Jie Ren et al. (2023). "Out-of-Distribution Detection and Selective Generation for Conditional Language Models". In: *ICLR 2023*.

Inside LLM: Hidden states

Intuition: the singular value spectrum of hidden representations should differ between prompts that lead to hallucinations and those that produce factual answers



Eigenvalues analysis

Idea: eigenvalues should capture the interaction in latent space between the different representations corresponding to hallucinated and truthful sequences.

$$\mathrm{U}_{\mathrm{EigVal}}(\mathbf{x}) = \log\det\left(\mathbf{\Sigma}^2(\mathbf{x})\right) = \sum_i \log\sigma_i^2.$$

Covariance matrices $\mathbf{\Sigma}(\mathbf{x})$ derived from:

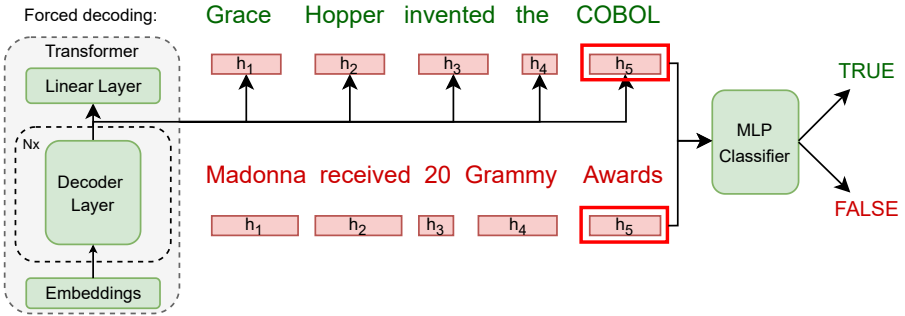
- sequence-level embeddings of $\mathbf{y}^i$ aggregated across sampled generations[24];
- token-level embeddings of the concatenation $\mathbf{x} \oplus \mathbf{y}$[25].

[24] Chao Chen et al. (2024). "INSIDE: LLMs' Internal States Retain the Power of Hallucination Detection". In: *ICLR 2024*.

[25] Gaurang Sriramanan et al. (2024b). "LLM-Check: Investigating Detection of Hallucinations in Large Language Models". In: *NeurIPS 2024*.

Statement accuracy prediction based on language model activations: SAPLMA

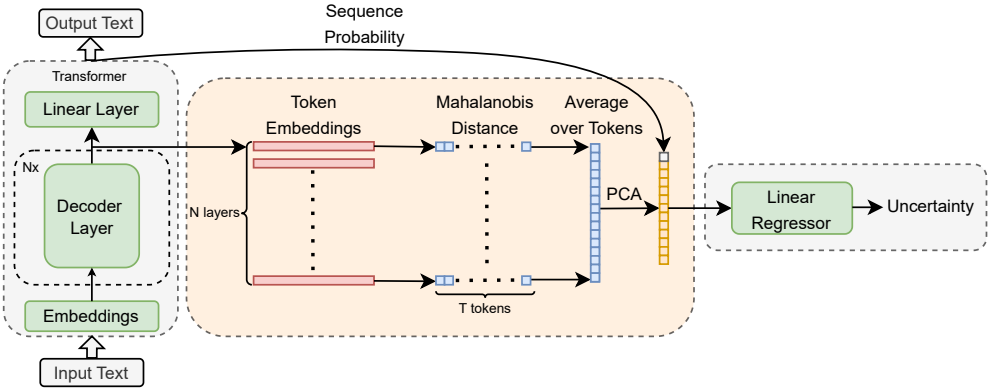
Idea: train on decoder layer activations to predict when LLM is uncertain[26].



[26] Amos Azaria and Tom Mitchell (2023). "The Internal State of an LLM Knows When It's Lying". In: *Findings of EMNLP 2023*.

Supervised average token-level relative Mahalanobis distance

Idea: aggregate token-level Mahalanobis distances to the cluster of "good" answers across all layers[27].



[27] Artem Vazhentsev et al. (2025a). "Token-Level Density-Based Uncertainty Quantification Methods for Eliciting Truthfulness of Large Language Models". In: *NAACL 2025*.

Problem of methods based on hidden states

- Other UQ methods based on hidden states:
  - Factoscope (He et al. 2024)
  - Span-level hallucination detection (Ma 2025)
  - MIND (Su et al. 2024)
  - "The Curious Case of Hallucinatory (Un)answerability" (Slobodkin et al. 2023)
  - ICR Probe (Zhang et al. 2025b)

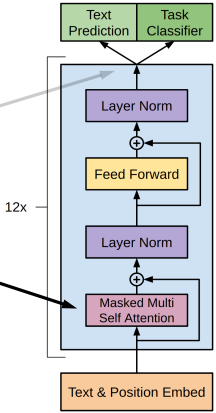
- All methods based on hidden states suffer from overfitting to a particular domain

4.2. Methods Based on Attention Weights

Inside LLM: Attention weights

Uncertainty quantification methods can leverage two **internal signals** from LLMs:

- Hidden states. The embedding vectors from each decoder layer for every generated token.
- Attention weights. The lower-triangular matrices, which illustrate how each token attends to the previous tokens during generation.



Transformer Decoder Architecture

Inside LLM: Attention weights

Intuition: the model attends differently when generating hallucinations versus truthful responses

- Analyze the eigenvalues of attention matrices
- Propagate uncertainty from preceding tokens using attention weights
- Train a supervised hallucination detector on attention-based features

Eigenvalues analysis of attention matrices

Idea: extend eigenvalue analysis[28] to the attention matrices A_{lh} at each layer $l \in L$ and head $h \in H$:

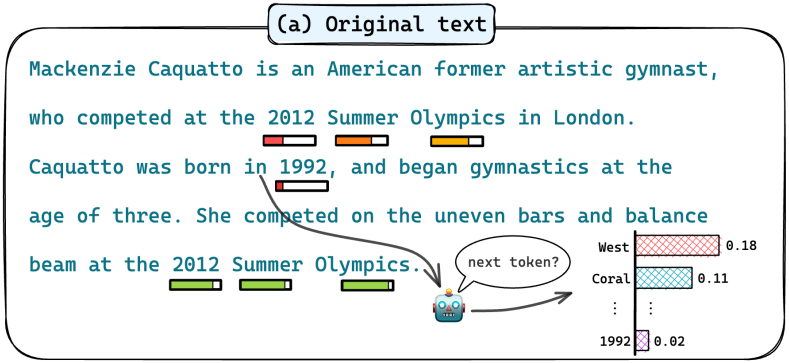
$$\mathrm{U}_{\mathrm{Att.Score}}(\mathbf{x}) = \sum_{h \in H} \log\det(\mathrm{A}_{lh}) = \sum_{h \in H} \sum_{j} \log \mathrm{A}_{lh}^{jj}.$$

Note: For lower-triangular attention matrices, the eigenvalues are equal to the diagonal elements, which correspond to attention on the last preceding token.

[28] Gaurang Sriramanan et al. (2024b). "LLM-Check: Investigating Detection of Hallucinations in Large Language Models". In: *NeurIPS 2024*.

Problem of conditional dependency of generation steps

Problem: conditional dependency on previous steps means the LLM assumes everything generated so far is correct, which may not always be the case.



Attention encodes conditional dependency of generation steps

Problem: LLMs provide the conditional probability distribution, assuming all previous tokens are correct:

$$P(y_i \mid \mathbf{y}_{<i}, \mathbf{x}).$$

We need the probability that does not depend on previously generated tokens:

$$P(y_i \mid \mathbf{x}).$$

Attention encodes conditional dependency of generation steps

Problem: LLMs provide the conditional probability distribution, assuming all previous tokens are correct:

$$P(y_i \mid \mathbf{y}_{<i}, \mathbf{x}).$$

We need the probability that does not depend on previously generated tokens:

$$P(y_i \mid \mathbf{x}).$$

Toy simplification (1-step dependency): assume $P(y_i \mid \mathbf{y}_{<i}) \approx P(y_i \mid y_{i-1})$ and tokens are binary (T/F):

$$\begin{aligned} P(y_i{=}\mathrm{T}) = {} & P(y_i{=}\mathrm{T} \mid y_{i-1}{=}\mathrm{T})\, P(y_{i-1}{=}\mathrm{T}) \\ & + P(y_i{=}\mathrm{T} \mid y_{i-1}{=}\mathrm{F})\big(1 - P(y_{i-1}{=}\mathrm{T})\big). \end{aligned}$$

Therefore: Cannot rely solely on raw LLM token probabilities.

Focus

Idea: use attention to propagate uncertainty from previous tokens into the reweighted token-level score $\hat{u}_i$[29].

$$\hat{u}_i = u_i + \alpha \cdot p_i,$$

Token-level uncertainty score,
e.g., the sum of the negative log probability and entropy

[29] Tianhang Zhang et al. (2023). "Enhancing Uncertainty-Based Hallucination Detection with Stronger Focus". In: *EMNLP 2023*.

Focus

Idea: use attention to propagate uncertainty from previous tokens into the reweighted token-level score $\hat{u}_i$[29].

$$\hat{u}_i = u_i + \alpha \cdot p_i,$$

Focus on the preceding tokens

where

- $\alpha = \gamma \cdot \mathbb{1}(y_i \in K)$ is a hyperparameter that depends on the set of keywords K,
- $p_i = \sum_{j=0}^{i-1} w_{i,j} \cdot \hat{u}_j$ is the weighted accumulated penalty, weights $w_{i,j}$ derived from attention between tokens y_i and y_j after max-pooling for all the layers and heads.

[29] Tianhang Zhang et al. (2023). "Enhancing Uncertainty-Based Hallucination Detection with Stronger Focus". In: *EMNLP 2023*.

Focus

- The sequence-level score is calculated by averaging the token-level scores:

$$\mathrm{U}_{\text{focus}} = \frac{1}{L} \sum_{i=1}^{L} \hat{u}_i,$$

where L is a length of output sequence $\mathbf{y}$.

[30] Tianhang Zhang et al. (2023). "Enhancing Uncertainty-Based Hallucination Detection with Stronger Focus". In: *EMNLP 2023*.

Focus

- The sequence-level score is calculated by averaging the token-level scores:

$$\mathrm{U}_{\text{focus}} = \frac{1}{L} \sum_{i=1}^{L} \hat{u}_i,$$

where L is a length of output sequence $\mathbf{y}$.

- The impact of the penalty p_i[30]:

Method	AUC-PR	Spearman Corr.
GPT-3 Perplexity	83.21	53.93
$\frac{1}{\lvert\mathbf{y}\rvert} \sum_{i=0}^{\lvert\mathbf{y}\rvert} u_i$	82.07	37.29
+ penalty p_i	**86.68**	**55.84**

[30] Tianhang Zhang et al. (2023). "Enhancing Uncertainty-Based Hallucination Detection with Stronger Focus". In: *EMNLP 2023*.

Recurrent attention-based uncertainty quantification: RAUQ

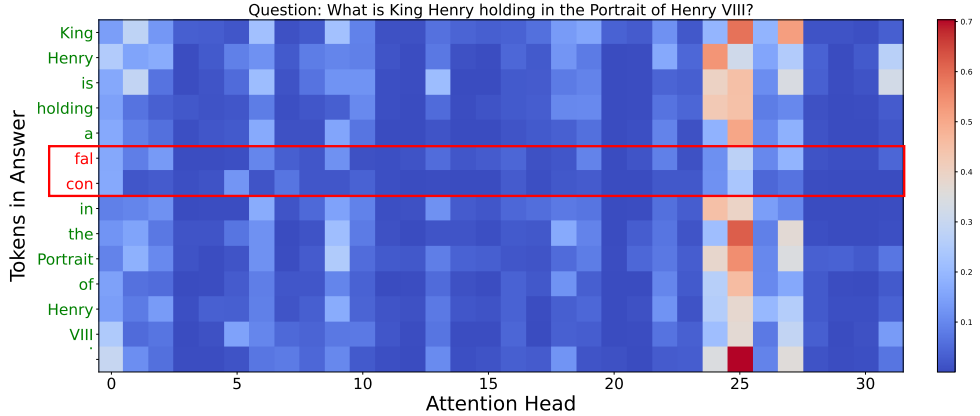
Idea: identify patterns in attention maps that reveal hallucinations[31].

- *Question*: What is King Henry holding in the Portrait of Henry VII?
- *Correct Answer*: gloves and dagger.
- *LLM Answer (Llama-3.1 8b)*: King Henry is holding a **falcon** in the Portrait of Henry VII.

[31] Artem Vazhentsev et al. (2025b). "Uncertainty-Aware Attention Heads: Efficient Unsupervised Uncertainty Quantification for LLMs". In: *arXiv preprint arXiv:2505.20045*.

Recurrent attention-based uncertainty quantification: RAUQ

- Most attention heads show low weights.
- The 25th head: high attention for correct tokens, low for the hallucinated token.



Recurrent attention-based uncertainty quantification: RAUQ

1. Select the most informative attention head per layer:

$$\mathbf{h}_l(\mathbf{y}) = \arg\max_{h=1\ldots H} \frac{1}{L-1} \sum_{i=2}^{L} a_{i,i-1}^{lh}.$$

Recurrent attention-based uncertainty quantification: RAUQ

1. Select the most informative attention head per layer:

$$\mathbf{h}_l(\mathbf{y}) = \arg\max_{h=1\ldots H} \frac{1}{L-1} \sum_{i=2}^{L} a_{i,i-1}^{lh}.$$

2. Compute token-level layer-wise recurrent confidence score:

$$\mathbf{c}_l(y_i) = \begin{cases} P(y_i \mid \mathbf{x}), & \text{if } i = 1, \\ \alpha \cdot P(y_i \mid \mathbf{y}_{<i}, \mathbf{x}) + (1-\alpha) \cdot a_{i,i-1}^{l\ \mathbf{h}_l} \cdot \mathbf{c}_l(y_{i-1}), & \text{if } i > 1. \end{cases}$$

Recurrent attention-based uncertainty quantification: RAUQ

1. Select the most informative attention head per layer:

$$\mathbf{h}_l(\mathbf{y}) = \arg\max_{h=1\ldots H} \frac{1}{L-1} \sum_{i=2}^{L} a_{i,i-1}^{lh}.$$

2. Compute token-level layer-wise recurrent confidence score:

$$\mathbf{c}_l(y_i) = \begin{cases} P(y_i \mid \mathbf{x}), & \text{if } i = 1, \\ \alpha \cdot P(y_i \mid \mathbf{y}_{<i}, \mathbf{x}) + (1-\alpha) \cdot a_{i,i-1}^{l\ \mathbf{h}_l} \cdot \mathbf{c}_l(y_{i-1}), & \text{if } i > 1. \end{cases}$$

3. Aggregate the token-level layer-wise uncertainty scores to the final score:

$$\mathrm{U}_{\mathsf{RAUQ}}(\mathbf{y}) = \max_{l \in \mathcal{L}} \Big[-\frac{1}{L} \sum_{i=1}^{L} \log \mathbf{c}_l(y_i) \Big].$$

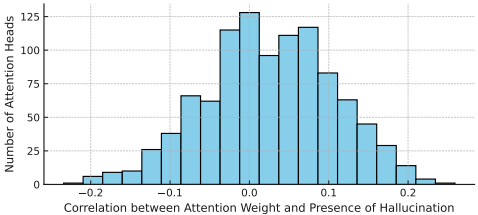
RAUQ: Preliminary results

UQ Method	Llama-3.1 8B			Gemma-3 12B			Mean
	QA	ATS	NMT	QA	ATS	NMT	
MSP	.353	.173	.395	.382	.317	.479	.350
Mean Token Entropy	.335	**.336**	.437	.447	.311	.576	.407
CCP	.322	.195	.343	.392	.338	.428	.336
Attention Score	.075	.077	.172	.061	.002	.126	.085
Focus	.282	.259	.372	.384	.310	.499	.351
Eigen Score	.363	.129	.394	.367	.086	.346	.281
Semantic Entropy	.339	.119	.393	.392	.048	.353	.274
SAR	**.377**	.214	.437	.406	.096	.458	.331
Semantic Density	.136	.316	.382	.139	**.353**	.405	.289
RAUQ	.354	.335	**.494**	**.447**	.325	**.613**	**.428**

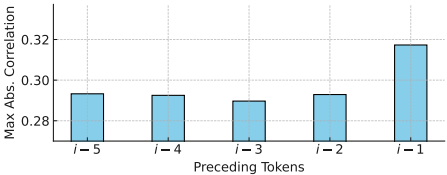
Table: Mean PRR↑ across tasks for the evaluated LLMs

- RAUQ shows the strongest overall performance, consistently ranking one of the best by average performance in a task.

Attention weights correlate with hallucination presence



(a) Correlation across various attention heads.

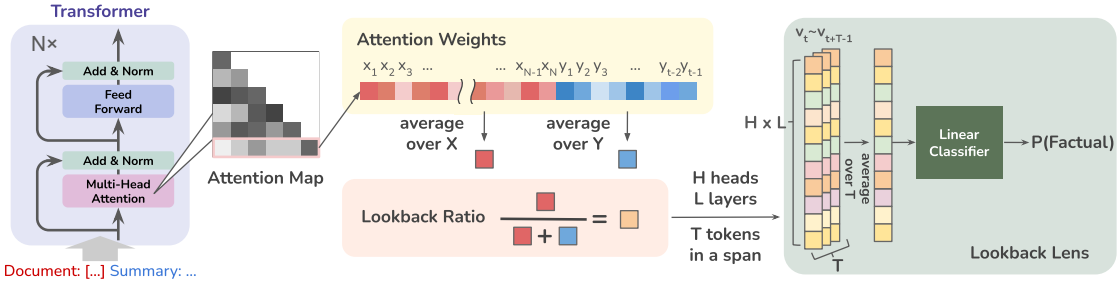


(b) Max absolute attention–hallucination correlation.

- **Left:** A small subset of uncertainty-aware attention heads exhibits a high correlation with hallucinations.
- **Right:** The most informative attention weight is the weight to the immediately preceding token.

Lookback lens

Idea: how much the model "looks back" at the provided context is a strong signal of factuality[32].



[32] Yung-Sung Chuang et al. (2024). "Lookback Lens: Detecting and Mitigating Contextual Hallucinations in Large Language Models Using Only Attention Maps". In: *EMNLP 2024*.

Trainable Attention-based conditional Dependency: TAD

Idea: attention implicitly encodes recurrent conditional dependency between generation steps, which we can learn[33].

$$\mathrm{U}(y_i) = U\big(P(y_i \mid \mathbf{y}_{<i}, \mathbf{x}), \boldsymbol{U}_{<i}, Att_i\big).$$

Estimation of the unconditional uncertainty for the i-th token

[33] Artem Vazhentsev et al. (2025c). "Unconditional Truthfulness: Learning Conditional Dependency for Uncertainty Quantification of Large Language Models". In: *EMNLP 2025*.

Trainable Attention-based conditional Dependency: TAD

Idea: attention implicitly encodes recurrent conditional dependency between generation steps, which we can learn[33].

$$\mathrm{U}(y_i) = U\big(P(y_i \mid \mathbf{y}_{<i}, \mathbf{x}), \boldsymbol{U}_{<i}, Att_i\big).$$

Linear regression

[33] Artem Vazhentsev et al. (2025c). "Unconditional Truthfulness: Learning Conditional Dependency for Uncertainty Quantification of Large Language Models". In: *EMNLP 2025*.

Trainable Attention-based conditional Dependency: TAD

Idea: attention implicitly encodes recurrent conditional dependency between generation steps, which we can learn[33].

$$\mathrm{U}(y_i) = U\big(P(y_i \mid \mathbf{y}_{<i}, \mathbf{x}), \boldsymbol{U}_{<i}, Att_i\big).$$

LLM token probability

[33] Artem Vazhentsev et al. (2025c). "Unconditional Truthfulness: Learning Conditional Dependency for Uncertainty Quantification of Large Language Models". In: *EMNLP 2025*.

Trainable Attention-based conditional Dependency: TAD

Idea: attention implicitly encodes recurrent conditional dependency between generation steps, which we can learn[33].

$$\mathrm{U}(y_i) = U\big(P(y_i \mid \mathbf{y}_{<i}, \mathbf{x}), \boldsymbol{U}_{<i}, Att_i\big).$$

Predicted uncertainty of the previous tokens

[33] Artem Vazhentsev et al. (2025c). "Unconditional Truthfulness: Learning Conditional Dependency for Uncertainty Quantification of Large Language Models". In: *EMNLP 2025*.

Trainable Attention-based conditional Dependency: TAD

Idea: attention implicitly encodes recurrent conditional dependency between generation steps, which we can learn[33].

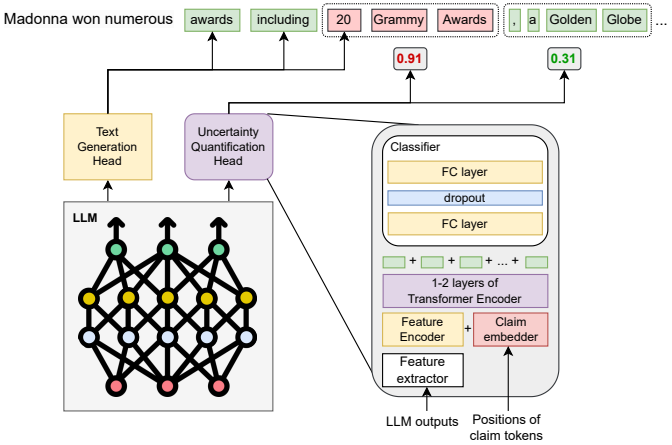
$$\mathrm{U}(y_i) = U\big(P(y_i \mid \mathbf{y}_{<i}, \mathbf{x}), \boldsymbol{U}_{<i}, \mathit{Att}_i\big).$$

Features derived from attention weights

- Att_i is a vector of LLM attention weights $a_{i,i-l}$ from the $(i-l)$-th token to the i-th token from all layers and heads, $l = 1, \ldots, i-1$.

[33] Artem Vazhentsev et al. (2025c). "Unconditional Truthfulness: Learning Conditional Dependency for Uncertainty Quantification of Large Language Models". In: *EMNLP 2025*.

Let's put things together: Pre-trained UQ heads[34]



Python library:

Hugging Face repo:

[34] Artem Shelmanov et al. (2025). "A Head to Predict and a Head to Question: Pre-trained Uncertainty Quantification Heads for Hallucination Detection in LLM Outputs". In: *EMNLP 2025*.

Performance of supervised methods compared to unsupervised I

	Method \ Domain	Biographies (in domain)	Cities	Movies	Inventions	Books	Artworks	Landmarks	Events
Unsup.	MSP	.412	.310	.205	.319	.145	.317	.135	.141
	Max Token Entropy	.416	.289	.241	.381	.171	.321	.141	.161
	CCP	.496	.368	.267	.380	.167	.382	.196	.171
Supervised	SAPLMA	.536	.435	.269	.350	.292	.534	**.350**	.235
	Lookback lens	.557	.449	.254	.391	.259	.464	.257	.295
	UHead (Ours)	**.660**	**.487**	**.466**	**.485**	**.395**	**.561**	.340	**.369**

Table: PR-AUC of claim-level hallucination detection. The supervised method is trained only on English *biographies* data.

- Performance gains for supervised UQ methods over unsupervised might be huge: up to 20 percentage points of PR-AUC.
- Supervised hallucination detectors might achieve good generalization to other domains: you can train on one domain and generalize to another.

Performance of supervised methods compared to unsupervised II

	Method \ Language	English (in domain)	Russian	Chinese	German
Unsup.	MSP	.180	.433	.307	.203
	Max Token Entropy	.202	.437	.444	.217
	CCP	.307	.493	.439	.306
Supervised	SAPLMA	.342	.514	.331	.391
	Lookback lens	.359	.576	.479	.390
	UHead (Ours)	**.457**	**.581**	**.556**	**.455**

Table: PR-AUC of claim-level hallucination detection. Supervised detectors were trained only on English *biographies* data.

- Supervised hallucination detectors achieve good generalization across languages: e.g. you can train on English and detect hallucinations in German generations.

Performance of supervised methods compared to unsupervised III

	Method \ Task	SamSum	CNN	WMT19	Mean PRR
	MSP (unsupervised)	**.298**	**.157**	**.569**	**.342**
Supervised	Factoscope	.077	.023	.131	.077
	SAPLMA	.045	.021	-.250	-.061
	MIND	.077	.048	.174	.099
	Sheeps	.104	-.021	.157	.080
	Lookback Lens	-.026	-.032	.018	-.013
	TAD (Ours)	.035	.003	.234	.091

Table: PRR for sequence-level selective generation. Supervised methods are trained on the QA task.

- Supervised methods generalize poorly across different tasks: you cannot train on QA and expect generalization to MT.

Example of using attention-based methods in LM-Polygraph

```
from transformers import AutoModel, AutoTokenizer
from lm_polygraph import estimate_uncertainty
from lm_polygraph.model_adapters import WhiteboxModel
from lm_polygraph.estimators import *

model_path = "meta-llama/Llama-3.1-8B-Instruct"
base_model = AutoModel.from_pretrained(model_path)
tokenizer = AutoTokenizer.from_pretrained(model_path)

ue_method = RAUQ()
input_text = "Who is George Bush?"
estimate_uncertainty(model, ue_method, input_text)
```

Any attention-based UQ method can be used with just one line

Example of using supervised UQ methods

```
from transformers import AutoModelForCausalLM, AutoTokenizer
from luh import AutoUncertaintyHead, CausalLMWithUncertainty

llm = AutoModelForCausalLM.from_pretrained(
    "google/gemma-2-9b-it")
tokenizer = AutoTokenizer.from_pretrained(
    "google/gemma-2-9b-it")
uhead = AutoUncertaintyHead.from_pretrained(
    "llm-uncertainty-head/uhead_gemma-2-9b-it", base_model=llm)
llm_adapter = CausalLMWithUncertainty(llm, uhead, tokenizer=tokenizer)

# tokenize text and prepare inputs ...
output = llm_adapter.generate(inputs)
# output = (generated tokens, uncertainty score)
```

Takeaways & QA

- Internal signals of LLMs offer valuable insights into the truthfulness of generated responses.
- Attention matrices are particularly effective for identifying potential hallucinations and better generalize compared to hidden states.
- Supervised UQ methods **drastically outperform** unsupervised methods for in domain tasks and **computationally efficient**
- Check out our uncertainty head framework and LM-Polygraph for implementations of supervised UQ methods
- In future work, we should scale up training data and seek for efficient integration of various sources of supervision.

5. Uncertainty Quantification for Visual Language Models

Lyudmila Rvanova

What is a vision–language model (VLM)?

- Vision–Language Models generate text conditioned on **both an image and a text prompt**
- The model must:
  - perceive visual content
  - align vision with language
  - generate a textual answer
- All of this happens within a single **autoregressive generation process**



Prompt: What is unusual in this image?

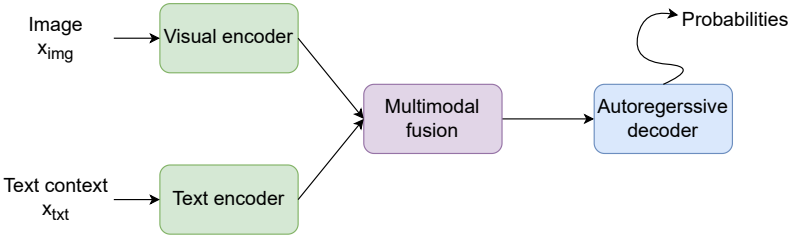
Model output: The bicycle has square wheels.

Visual language models: definition

- VLMs generate text autoregressively, but conditioning happens on latent multimodal representations:

$$p\big(y \mid x^{\mathrm{img}}, x^{\mathrm{txt}}\big) = \prod_{t=1}^{T} p\big(y_t \mid y_{<t}, x^{\mathrm{img}}, x^{\mathrm{txt}}\big).$$

- This distribution is computed **after multimodal fusion**



What becomes observable only in vision–language models

- In VLMs, uncertainty is no longer purely **linguistic**
- Model confidence must reflect whether a claim is **visually supported**, not only textually likely
- Identical or fluent answers may rely on **different visual evidence** — or none at all
- Strong language priors can dominate even when **visual evidence is weak or absent**

Visual representational uncertainty

- In vision–language models, uncertainty can arise **within the visual representation itself**, before grounding or text generation
- This occurs when the visual encoder:
  - cannot uniquely determine a visual interpretation
  - produces unstable representations under small image perturbations
  - fails to encode task-relevant visual information
- Such uncertainty may remain **invisible at the output level**, even when generated answers appear fluent and consistent
- Operationally, small Δx^{img} can induce large changes in z, indicating a lack of representational stability

Epistemic uncertainty of visual tokens

- Epistemic uncertainty can be operationalized as **instability of visual token representations** under small perturbations of the input image[35]
- A single LVLM with a fixed vision encoder encodes both the original image x^{img} and a slightly perturbed image $x^{\text{img}} + \delta$
- If a visual token z_m changes significantly,

$$\left\| z_m(x^{\text{img}} + \delta) - z_m(x^{\text{img}}) \right\| \text{ is large,}$$

this indicates **high epistemic uncertainty** about the corresponding visual region

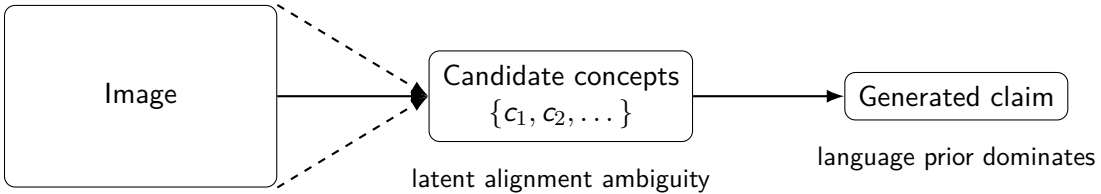
[35] Hoigi Seo et al. (2025). "On Epistemic Uncertainty of Visual Tokens for Object Hallucinations in Large Vision-Language Models". In: *NeurIPS 2025*.

Vision–language alignment uncertainty

- **Vision–language alignment** determines how visual representations constrain text generation
- Alignment uncertainty arises when visual evidence is:
  - weak or ambiguous
  - distributed across multiple plausible regions
  - insufficient to disambiguate competing interpretations
- In such cases, generation may rely primarily on **language priors rather than visual evidence**

Vision–language alignment uncertainty

- When alignment uncertainty is high, generation becomes weakly constrained by visual evidence
- This results in **fluent and confident**, yet **visually ungrounded outputs**, which serve as an observable signature of alignment uncertainty[36]



Interpretation: alignment uncertainty is not directly observable, but manifests through unstable or ungrounded generation behavior.

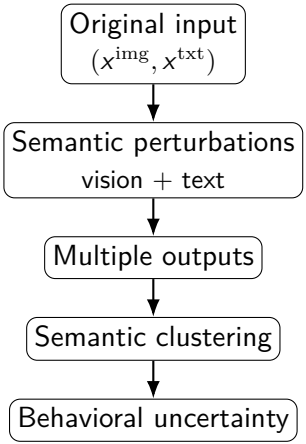
[36] Yifan Li et al. (2023). "Evaluating Object Hallucination in Large Vision-Language Models". In: *EMNLP*.

Quantifying uncertainty through behavior

- Given multiple outputs under semantic-preserving perturbations, answers are clustered by meaning[a]
- Behavioral uncertainty is quantified as entropy over semantic clusters:

$$H_{\text{beh}} = -\sum_{k} p(c_k) \log p(c_k)$$

- Higher entropy indicates greater semantic instability



[a] Sushant Gautam, Michael A. Riegler, and Pål Halvorsen (2025). "HEDGE: Hallucination Estimation via Dense Geometric Entropy for VQA with Vision-Language Models". In: *arXiv preprint arXiv:2511.12693*.

Dropout Decoding at the Alignment Boundary

$$\bar{q}(y) = \frac{1}{N}\sum_{i=1}^{N} q_i(y), \qquad U_{\text{epi}}(v_i) = D_{\text{KL}}\big(q_i(y) \,\|\, \bar{q}(y)\big)$$

$$v_i' = \begin{cases} \varnothing, & U_{\text{epi}}(v_i) > \tau \\ v_i, & \text{otherwise} \end{cases} \qquad z' = f_{\text{fusion}}(v_1', \ldots, v_N', t)$$

- $q_i(y)$ — text distribution induced by visual token v_i
- High $U_{\text{epi}}(v_i)$ indicates unreliable vision–language alignment
- Uncertain visual tokens are suppressed before decoding

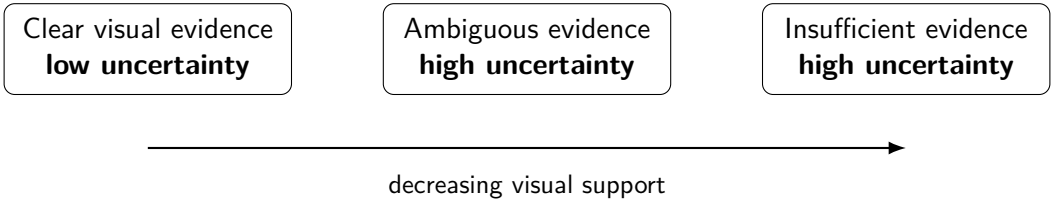
[37] Yixiong Fang et al. (2025). "Enhancing Vision-Language Model Reliability with Uncertainty-Guided Dropout Decoding". In: *The Thirty-ninth Annual Conference on Neural Information Processing Systems*.

Residual uncertainty

- After visual evidence and vision–language alignment are fixed, uncertainty in text generation still remains
- This uncertainty originates in the **autoregressive decoder**, as in text-only LLMs
- **In vision–language models,** decoder-side signals capture only the **residual uncertainty**
- They do not reflect:
  - visual perception uncertainty
  - vision–language alignment failures

Evaluating uncertainty awareness in vision–language models

- Beyond estimating uncertainty signals, the goal is to assess whether models are uncertain **in the appropriate situations**
- Given the same question, does the model change its confidence when the visual evidence is weakened or removed?



38

[38] Khyathi Chandu et al. (2025). "CertainlyUncertain: A Benchmark and Metric for Multimodal Epistemic and Aleatoric Awareness". In: *ICLR*.

Factorizing uncertainty sources via controlled perturbations

- Analyzing predictive uncertainty of the output y under **controlled variations of the input**[39] x:

$$H[y \mid x, w] = U_{\text{attr}} + U_{\text{rephrase}} + U_{\text{residual}},$$

where

- U_{attr} — uncertainty due to **which visual evidence is used**
  - variation in objects, regions, or attributes serving as evidence
- U_{rephrase} — uncertainty due to **linguistic formulation**, with the visual scene fixed
  - semantically equivalent rephrasings of the same query
- U_{residual} — remaining uncertainty once visual attribution and wording are controlled

[39] Ziran Yang et al. (2025b). "Understanding the Sources of Uncertainty for Large Language and Multimodal Models". In: *ICLR Workshop: QUESTION*.

Preliminary results: Experiments with LVLM on PaliGemma

Table: PRR@0.5 (normalized), computed using AlignScore as the quality metric.

UQ Method	MMStar	MathVista	iNat	POPE-VQA
CCP	0.418	0.040	0.045	-0.042
DegMat_Jaccard_score	0.310	-0.042	-0.141	-0.037
DegMat_NLI_score_contra	-0.046	-0.046	-0.195	-0.067
Eccentricity_NLI_score_entail	0.030	0.096	0.067	0.067
EigenScore	0.305	-0.007	-0.065	-0.025
Entropy	0.184	0.071	0.068	0.069
MahalanobisDistanceSeq	-0.118	0.252	0.231	0.146
MaximumSequenceProbability	0.025	0.025	0.024	0.024
MonteCarloSequenceEntropy	0.230	0.106	0.078	0.071
PPLRMDSeq	-0.176	-0.221	-0.260	-0.202
RDESeq	0.250	-0.164	-0.326	-0.271
RelativeMahalanobisDistanceSeq	-0.294	0.137	0.197	0.150
SemanticEntropy	0.197	0.115	0.083	0.078
TokenSAR	-0.384	-0.181	-0.228	-0.170

PRR@0.5↑ (Normalized), computed using AlignScore as the quality metric.

Takeaways & QA

- Uncertainty in VLMs is **multi-source**:
  - visual representation
  - vision–language alignment
  - linguistic variability
  - residual decoder uncertainty
- Meaningful uncertainty requires **localizing its source** within the VLM pipeline
- What matters is not confidence itself, but whether uncertainty tracks visual support.

6. Uncertainty Normalization

Roman Vashurin

Why you should normalize uncertainty?

- Raw uncertainty scores effective for ranking model outputs, but they are hard to interpret by end users:

$$\mathrm{C}_{\mathrm{MTE}} = 6.3$$

- Normalization helps translate uncertainty into interpretable confidence scores (bounded [0,1]).

- To be meaningful, uncertainty scores should be calibrated: approximate the generation quality of LLM.

Normalization approaches

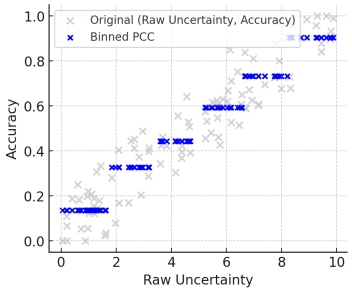
- **Linear Scaling**
  - Scales raw uncertainty linearly to [0,1] based on min/max values.
- **Quantile Scaling**
  - Uses quantiles to map uncertainty scores to [0,1], sensitive to data distribution.
- **Performance-Calibrated Confidence (PCC)**
  - Directly links confidence scores with expected model performance.
  - Two types: Binned PCC and Isotonic PCC[40].

[40] Roman Vashurin et al. (2025a). "Benchmarking uncertainty quantification methods for large language models with lm-polygraph". In: *TACL 2025*.

Performance-Calibrated Confidence: PCC

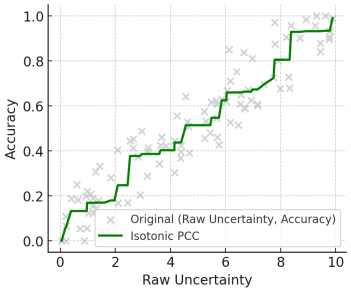
Binned PCC

- Calibrated uncertainty = mean quality score in the uncertainty bin.
- Potential ordering discrepancies.

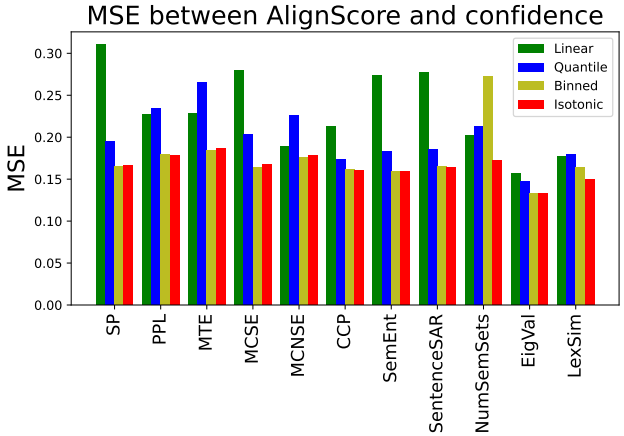


Isotonic PCC

- Fits isotonic regression between uncertainty and quality scores.
- Maintains ordering.



Normalization results: Confidence calibration



Isotonic PCC consistently outperforms simpler normalization approaches (Linear, Quantile) in calibration quality.

Example of using normalization in LM-Polygraph I

- Firstly we need to fit the normalizers on held-out data. This can be achieved by executing `polygraph_normalize`:

```
HYDRA_CONFIG=$CONFIG_PATH polygraph_normalize save_path="./"
↪  man_paths=$TRAIN_DATA_PATH
```

- Here `CONFIG_PATH` corresponds to the location of the YAML config and `TRAIN_DATA_PATH` is the path to the results of running ‘polygraph_eval‘ on the held-out dataset.

Example of using normalization in LM-Polygraph II

- Now we can load normalizers and obtain calibrated scores:

```
import pickle
from lm_polygraph.normalizers.isotonic_pcc import
    ↪ IsotonicPCCNormalizer

# Load saved fitted normalizers, can be obtained by running
    ↪ `polygraph_normalize`
with open('fitted_normalizers.json', 'rb') as f:
    normalizers = pickle.load(f)

normalizer = IsotonicPCCNormalizer.loads(normalizers[("AlignScore",
    ↪ "CocoaMSP", "isotonic_pcc")])
calibrated_confidence = normalizer.transform([-5.0])
```

Takeaways & QA

- Confidence normalization is crucial for interpretability by end-users.
- Isotonic PCC provides best balance: retains performance and provides meaningful confidence scores.
- Simple normalization methods useful but inferior to PCC-based methods.
- Recommended: Isotonic PCC for practical deployments.

7. Benchmarking Uncertainty Quantification Methods

Roman Vashurin

Tasks for evaluation of UQ methods

Selective generation (predict only when certain)

- Uncertainty score for entire output.
- Best suited for short-form responses.



What is the capital of Austria?



The capital of Austria is Salzburg.

Uncertainty: **0.98**

Claim-level hallucination detection

- Uncertainty score per extracted claim.
- More fine-grained, but limited to UQ methods that support span-level calculations.



What Nobel prizes did Marie Curie win?



Marie Curie won the Nobel Prize in **Chemistry in 1910** and **Physics in 1903**.

Selective generation: Example



What is the capital of Austria?



The capital of Austria is **Salzburg**. ← **Uncertainty:** 98%

Reference Answer: Vienna

⇒ **Quality Metric** (Output vs Reference Similarity): low

⇒ Expect high uncertainty

Figure: Example of an incorrect LLM answer with high uncertainty.

Selective generation: Ranking task

What do ranking metrics measure?

- Evaluate how well uncertainty scores **rank** model outputs by quality.
- We measure **relative order**, not the actual uncertainty values.
- If you are interested in absolute values (e.g., "is 0.9 truly uncertain?"), that is a calibration task (e.g., Expected Calibration Error: ECE).

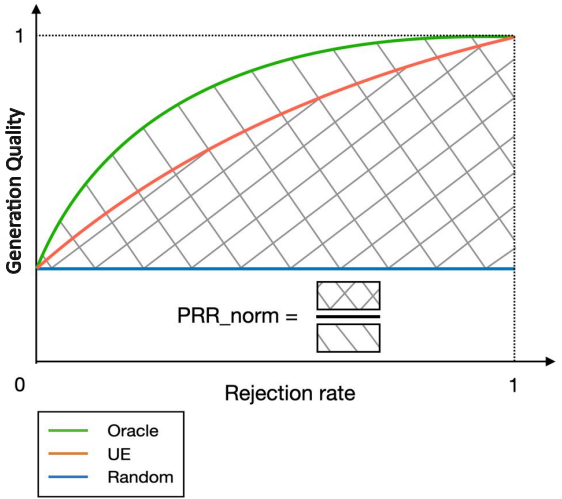
Selective generation: Ranking metrics

- **Spearman correlation** — measures monotonic relationship between uncertainty estimates and quality metrics.
- **Reversed Pairs Proportion (RPP)** — fraction of pairwise misorderings $u_1 < u_2$ but $q_1 < q_2$.
- Any binary metrics (**ROC-AUC** / **PR-AUC**) for a thresholded quality metric.
- **Prediction-Rejection Ratio (PRR)** — specifically designed for selective generation.

Prediction-rejection curve

Prediction–rejection curve: at rejection rate k:

- Reject top-k% most uncertain outputs
- Plot performance (average quality metric) on the retained subset

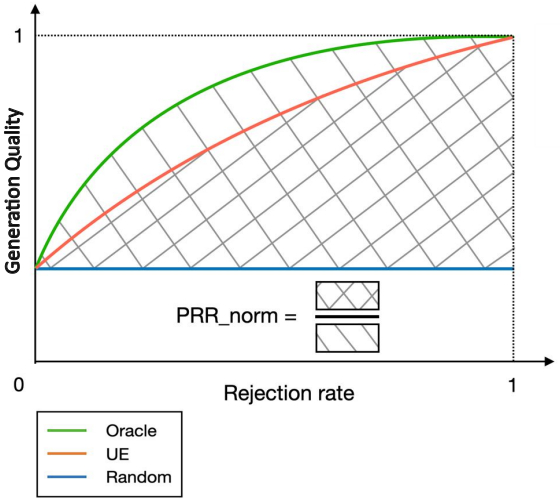


Prediction-Rejection Ratio: PRR

$$\mathbf{PRR_{norm}} = \frac{\mathrm{AUC_{unc}} - \mathrm{AUC_{rnd}}}{\mathrm{AUC_{oracle}} - \mathrm{AUC_{rnd}}}$$

- $\mathrm{AUC_{unc}}$: area under model's rejection curve
- $\mathrm{AUC_{rnd}}$: random rejection baseline
- $\mathrm{AUC_{oracle}}$: ideal ranking (worst-quality first)

Set **maximum rejection limit** (e.g. 50%) to avoid noisy tails.



Selective generation: Generation quality metrics I

- **Quality metrics** compare output to a gold reference (Malinin and Gales 2021).
- Quality values are later compared with uncertainty estimates with ranking metric.
- Common metrics: ROUGE-L, AlignScore, BERTScore, etc.


Who is the president of France?



Emmanuel Macron is the current president.

Uncertainty: 0.12
Gold reference: Emmanuel Macron
AlignScore: 0.99 $\Rightarrow$ Expect low uncertainty

Selective generation: Generation quality metrics II

Category	Quality Metrics	Scoring Mechanism	Features
Lexical Overlap	**ROUGE-L** (Lin 2004) **BLEU** (Papineni et al. 2002)	Exact n-gram matches	Cheap but poor correlation with semantics
Embedding-based	**BERTScore** (Zhang et al. 2020) **Sentence-BERT** (Reimers and Gurevych 2019)	Distance between pretrained embeddings	More robust to paraphrasing
Model-based Scoring	**COMET** (Rei et al. 2020) **AlignScore**(Zha et al. 2023)	Trained to predict information alignment	Conditioned on input question

Claim-level hallucination detection: Example



What Nobel prizes did Marie Curie win?



Marie Curie won the Nobel Prize in Chemistry in 1910 and Physics in 1903.

Extracted Claims:

1. Chemistry Nobel in 1910
   Uncertainty: 0.82
   Label: False
2. Physics Nobel in 1903
   Uncertainty: 0.18
   Label: True

Claim-level hallucination detection: Ranking metrics

- **ROC-AUC**: measures separability of True vs. False samples.
- **PR-AUC (False = Positive)**: focuses on flagging hallucinations. Better for imbalanced data.
- **Prediction-Rejection Ratio**

7.1. Experiments: Sequence-level Selective Generation[41]

[41] Experimental setup and some results discussed in this section are presented in: Roman Vashurin et al. (2025a). "Benchmarking uncertainty quantification methods for large language models with lm-polygraph". In: *TACL 2025*.

Experimental setup: Datasets & LLMs

- **Datasets for selective generation evaluation:**

Dataset	Type	Num. Instances train/test	Evaluated Subset	Avg. Document Len.	Avg. Target Len.	Language
CoQA	QA	7,199 / 500	2000	405.9	4	English
TriviaQA	QA	138,384 / 17,210	2000	18.8	4.3	English
MMLU	QA	99,842 / 14,042	5700	64.9	3	English
GSM8k	QA	7,473 / 1,319	1319	64.2	128.6	English
XSum	ATS	204,045 / 11,334	2000	544.2	30.4	English
WMT'14	NMT	40.8M / 3,003	2000	49.3	32.9	Fr.-to-Eng.
WMT'19	NMT	34.8M / 2,998	2000	52.5	33.5	Ger.-to-Eng.

- **LLMs:**
  - White-box base: Llama-3.1 8B Instruct, Gemma-3 12B.
  - White-box instruct: StableLM v2 12B Chat, Mistral v0.2 7B Instruct.
  - Black-box: GPT-4o Mini.

Results: QA with base LLMs

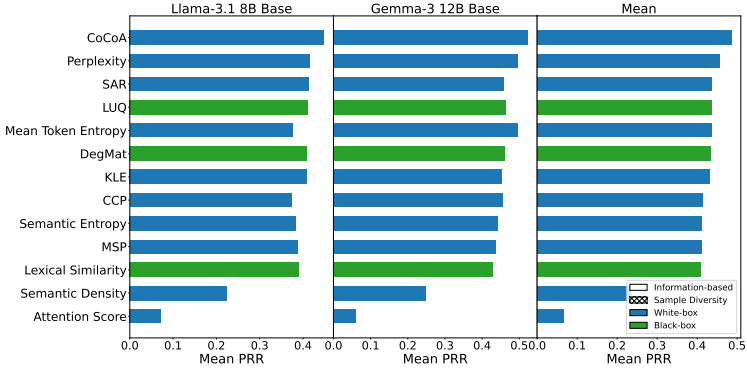


Figure: Mean PRR↑ aggregated over all QA datasets for each LLM.

- Best performance for QA: **CoCoA** and **Perplexity**.

Results: MT & summarization with base LLMs

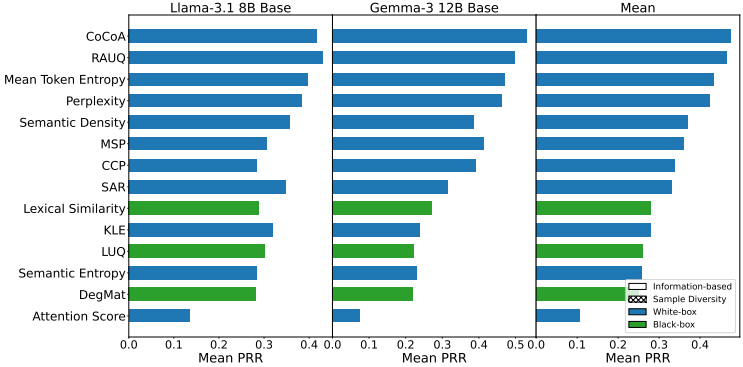


Figure: Mean PRR↑ aggregated over all MT and summarization datasets for each LLM.

- Best performance for long generation tasks is obtained by **CoCoA** and **RAUQ**.

Results: QA with black-box instruction-following LLMs

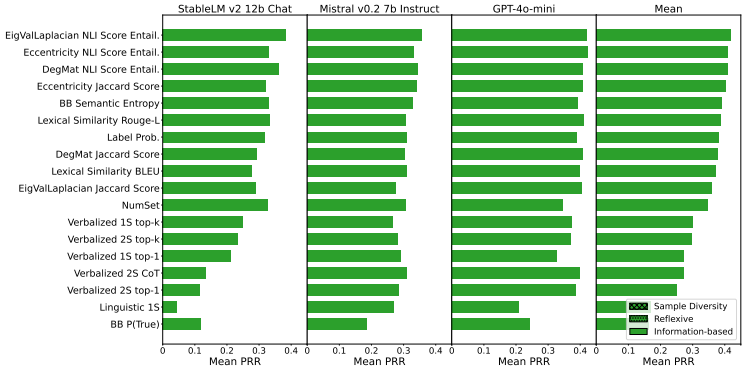


Figure: Mean PRR↑ aggregated over all selective QA tasks for each black-box LLM separately.

- Best performance for QA: **consistency-based methods**.
- **Verbalized** methods competitive only for advanced models, such as GPT-4o-mini.

7.2. Experiments: Claim-level Hallucination Detection

Experimental setup: Datasets, LLMs, metrics

- **Datasets:**
  - biographies of famous persons (input: `Tell me a bio of {person}`)
  - multi-lingual biographies: German, Russian, Chinese, Arabic
  - multi-domain: cities, movies, inventions, books
- **Evaluation pipeline:** GPT-4o as both splitter and fact-checker.
- **LLMs:**
  - Gemma 3 4B Instruct, Gemma 3 12B Instruct
  - Llama 3.1 8B, Llama 3.1 8B Instruct
- **Metric:** PR-AUC (hallucination is a positive class), PRR (0.5 max rejection).

Results: English biographies

UQ Method	Gemma-3-4b-it		Gemma-3-12b-it		Llama-3.1-8B		Llama-3.1-8B-Instruct	
	PR-AUC	PRR	PR-AUC	PRR	PR-AUC	PRR	PR-AUC	PRR
Maximum Probability	0.303	0.146	0.149	0.122	0.213	0.088	0.224	0.278
Perplexity	0.275	0.083	0.140	0.100	0.192	0.073	0.219	0.274
Max Token Entropy	0.321	0.180	0.164	0.155	0.184	0.049	0.253	0.296
P(True)	0.318	0.141	0.196	0.119	0.233	0.135	0.240	0.326
CCP	0.447	0.358	0.206	0.288	0.326	0.271	0.326	0.352
Attention Score	0.303	0.119	0.129	0.052	0.204	0.087	0.142	0.027
Focus	0.318	0.195	0.161	0.145	0.199	0.082	0.226	0.279
Frequency Scoring	0.520	0.502	0.215	0.337	0.292	0.309	0.308	0.401
TokenSAR	0.306	0.154	0.153	0.134	0.211	0.089	0.222	0.279
CoCoA (Freq. x CCP)	0.518	0.475	0.230	0.341	0.365	0.350	0.395	0.449

PR-AUC↑ and PRR↑ on the claim-level English biographies.

Top method: **CoCoA** combining **FrequencyScoring** and **CCP**.

7.3. How do I Evaluate My Novel UQ Method?

Benchmarking in LM-Polygraph

Run benchmarking in one line with LM-Polygraph:

```
polygraph_eval --config-name=config.yaml
```

Predefined configs available in `examples/configs`:

- **Sequence-level:** CoQA, TriviaQA, XSum, WMT, GSM8k, MMLU, ...
- **Claim-level:** Biographies in English, Russian, Arabic, Chinese, German.

Benchmarking in LM-Polygraph

Custom configs let you control:

- Dataset, LLM, and custom new UQ methods
- Quality and ranking metrics
- Parameters for prompting and UQ methods, seeds

Takeaways & QA

- **Two UQ tasks:** Sequence-level (e.g., QA, summarization) and claim-level (hallucination detection) benchmarks introduced.
- **Evaluation:** Ranking-based metrics (PRR, Spearman, ROC-AUC) and generation quality metrics (e.g., AlignScore, ROUGE-L).
- **Benchmarking toolkit:** `LM-Polygraph` enables reproducible benchmarking with support for both predefined and custom UQ methods.

8. Uncertainty for Reasoning and Agentic AI

Artem Shelmanov

Goals of introducing uncertainty in reasoning and agentic AI

- Improve final performance in complex tasks
- Reduce compute / token cost of reasoning or agentic workflow

What is reasoning?

- LLMs solve complex tasks by generating Chain-of-Thought (CoT) reasoning traces
- Given an input question $\mathbf{x}$, the model generates a sequence of reasoning steps

$$\mathbf{r} = \{r_1, r_2, \ldots, r_T\}$$

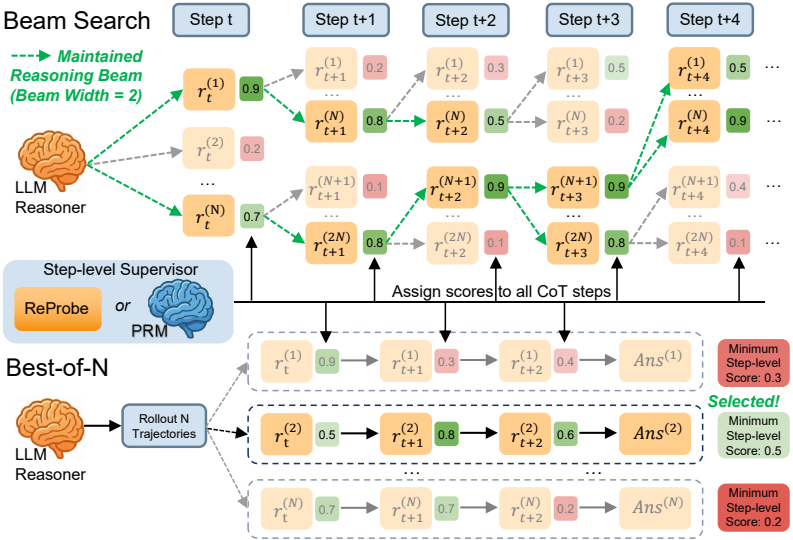
followed by a final answer

$$y = g(\mathbf{r}, \mathbf{x}).$$

- Formally, an LLM parameterized by θ models the joint distribution:

$$P_\theta(y, \mathbf{r} \mid \mathbf{x}) = \prod_{t=1}^{T} P_\theta(r_t \mid \mathbf{x}, \mathbf{r}_{<t}) \cdot P_\theta(y \mid \mathbf{x}, \mathbf{r})$$

Test time scaling (TTS) for reasoning



Idea 1: Uncertainty as a reasoning step scorer

- Use uncertainty to score the quality of reasoning steps and entire trajectories
- Scorer for a reasoning step is confidence:

$$Q(r_t) = C(r_t) = 1 - U(r_t)$$

- Scorer for the entire reasoning trajectory is aggregation of step-level scores (e.g. average):

$$Q(r_{1:T}) = \frac{1}{T} \sum_{t=1}^{T} C(r_t)$$

Deep think with confidence[42]

- **Strategy:** best of N with weighted majority voting
- Reasoning step is a **sliding window** of tokens
- **Scorers:** (1) avg. confidence in bottom 10% steps W_{b10}; (2) avg. confidence of steps in the tail window W_{tail}:

$$\text{TokConf}_i = -\frac{1}{|\text{Vocab}|}\sum_{j=1}^{|\text{Vocab}|}\log P_i(j|x, y_{<t}) \qquad Q_{\text{bottom-10}}(r_{1:T}) = \frac{1}{|W_{\text{b10}}|}\sum_{r_t \in W_{\text{b10}}} C_{r_t}$$

$$C_{r_t} = \frac{1}{|r_t|}\sum_{i \in r_t}\text{TokConf}_i \qquad Q_{\text{tail}}(r_{1:T}) = \frac{1}{|W_{\text{tail}}|}\sum_{t \in W_{\text{tail}}} C_t$$

- **Results:** outperform basic majority voting by up to +7.8 pp essentially for free

[42] Yichao Fu et al. (2025a). "Deep Think with Confidence". In: *Workshop on Eff. Reasoning at NeurIPS 2025*. Spotlight.

Scoring the entire reasoning trajectory with uncertainty[43]

- **Strategy:** Best of N with selection by lowest uncertainty; can be combined with weighted voting
- Uses **entire trajectory** without splitting into reasoning steps
- **Scorer** is cross-entropy with uniform distribution:

$$Q(r_{1:T}) = -\frac{1}{T \cdot |\text{Vocab}|} \sum_{t=1}^{T} \sum_{j=1}^{|\text{Vocab}|} \log P(j \mid x, y_{<t})$$

- **Results:** combined with weighted voting can outperform PRMs and ORMs

[43] Zhewei Kang, Xuandong Zhao, and Dawn Song (2025). "Scalable Best-of-N Selection for Large Language Models via Self-Certainty". In: *NeurIPS-2025*.

(Ours) Verification of reasoning steps via ReProbes[44]

- We adapt UHead for reasoning step scoring:
  ReProbe – a small neural network that uses LLMs internal states (hidden states, attention maps) as features to assess the quality of steps
- ReProbe can be viewed as a **supervised confidence estimator** for reasoning steps
- Introspection and Transformer architecture allows ReProbe to be very effective and efficient

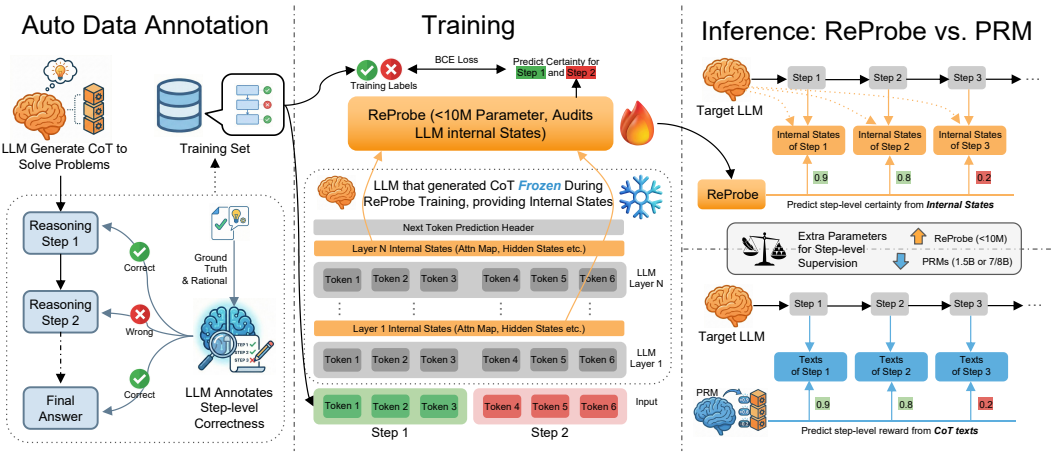
[44] Jingwei Ni et al. (2025). *Efficient Test-Time Scaling of Multi-Step Reasoning by Probing Internal States of Large Language Models*. arXiv preprint.

(Ours) Verification of reasoning steps via ReProbes[44]

- We adapt UHead for reasoning step scoring:
  ReProbe – a small neural network that uses LLMs internal states (hidden states, attention maps) as features to assess the quality of steps
- ReProbe can be viewed as a **supervised confidence estimator** for reasoning steps
- Introspection and Transformer architecture allows ReProbe to be very effective and efficient
- **Strategies:** Best of N, beam search (tree of thought with BFS)
- Reasoning steps are extracted via pattern matching

[44] Jingwei Ni et al. (2025). *Efficient Test-Time Scaling of Multi-Step Reasoning by Probing Internal States of Large Language Models*. arXiv preprint.

(Ours) ReProbe as an alternative to PRMs



(Ours) ReProbe preliminary results - I

Method	# Sample	Math (ID)			Planning (OOD)			QA (OOD)		Average		
		MATH	GSM8k	ProofNet	Trips	Meetings	Calendar	StrQA	SciQA	ID	OOD	Overall
Unsupervised Uncertainty Quantification (UQ)												
MaxProb	-	.221	.106	.194	.578	.655	.483	.114	.259	.174	.418	.326
MaxEntropy	-	.212	.122	.185	.545	.618	.443	.119	.227	.173	.390	.309
Perplexity	-	.205	.099	.176	.519	.572	.418	.110	.228	.160	.369	.291
P(True)	-	.164	.059	.172	.535	.608	.490	.126	.263	.132	.404	.302
Semantic Entropy	-	.257	.116	.173	.565	.610	.492	.111	.265	.182	.409	.324
Degree Matrix	-	.227	.089	.147	.534	.597	.484	.107	.258	.154	.396	.305
*PRMs **150**× Larger than ReProbes*												
Skywork-PRM-1.5B	Unk	.283	.412	.147	.433	.532	.502	.254	.408	.281	.426	.371
H4-Qwen2.5-PRM-1.5B-0.2	369K	.259	.171	.159	.597	.633	.495	.213	.228	.196	.433	.344
*PRMs **750**× to **810**× Larger than ReProbes*												
RLHFlow-PRM-Deepseek-8B	253K	.289	.540	.136	.583	.579	.504	.390	.518	.322	.515	.442
Universal-PRM-Qwen2.5-Math-7B	690K	.534	.624	**.329**	.730	.753	.691	.328	.330	.496	.566	.540
Qwen2.5-Math-7B-PRM800k	265K	**.586**	.613	.301	.708	.768	.727	.362	.404	.500	.594	.559
Qwen2.5-Math-PRM-7B	860K	.531	**.702**	.310	.711	.757	.745	.334	.429	**.514**	.595	.565
Reasoning Probes (ReProbes)												
★ ReProbe, Attn+Logit, Self-anno (Ours)	**32K**	.529	.594	.260	.735	.779	.779	.394	.404	.461	.618	.559
★ ReProbe, Attn+Logit, DeepSeek-anno (Ours)	**32K**	.465	.616	.243	.740	.802	.786	**.395**	.361	.441	.617	.551
★ ReProbe, Hidden States, Self-anno (Ours)	**32K**	.558	.673	.264	**.799**	**.819**	.785	.381	**.553**	.498	**.667**	**.604**
★ ReProbe, Hidden States, DeepSeek-anno (Ours)	**32K**	.534	.650	.281	.793	.817	**.789**	.344	.450	.488	.639	.582

Table: PR-AUC↑ for detecting incorrect reasoning steps (Qwen3-8B).

Results: ReProbes outperform unsupervised UQ methods and the best PRMs on OOD

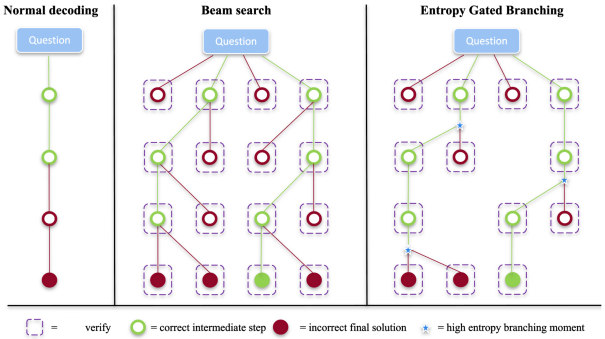
(Ours) ReProbe preliminary results - II

Method	# Sample	Math (ID)			Planning (OOD)		QA (OOD)		Average		
		MATH	GSM8k	ProofNet	Trips	Calendar	StrQA	SciQA	ID	OOD	Overall
PRMs 750× to 810× Larger than ReProbes											
Qwen3-8B pass@1 (Lower Bound)	-	92.4	95.6	74.1	8.1	23.5	86.8	92.7	87.4	43.3	59.8
Qwen3-8B pass@N (Upper Bound)	-	99.3	99.2	97.8	36.2	60.5	98.3	99.3	98.8	56.1	72.1
Qwen3-14B pass@1	-	93.4	97.6	76.0	38.7	40.5	91.9	95.6	89.0	54.3	67.3
Qwen2.5-Math-7B-PRM800k	263K	89.8	80.4	95.2	13.1	45.0	86.7	91.2	88.5	59.0	71.6
Qwen2.5-Math-PRM-7B	860K	88.1	95.4	93.6	13.1	41.5	79.2	83.6	92.4	54.4	70.7
Reasoning Probes (ReProbes)											
★ ReProbe, Attn+Logit, Self-anno (Ours)	**32K**	**90.3**	95.4	95.1	15.0	41.5	94.8	96.2	93.6	61.9	75.5
★ ReProbe, Attn+Logit, DeepSeek-anno (Ours)	**32K**	81.8	93.3	79.1	11.8	42.4	90.7	90.8	84.7	58.9	70.0
★ ReProbe, Hidden States, Self-anno (Ours)	**32K**	84.1	97.3	90.6	**15.6**	40.5	91.2	92.8	90.7	60.0	73.2
★ ReProbe, Hidden States, DeepSeek-anno (Ours)	**32K**	86.8	**98.8**	**95.6**	13.4	**48.0**	**96.7**	**96.7**	**93.7**	**63.7**	**76.6**

Table: Beam search decoding accuracy across datasets (Qwen3-8B).

Results: ReProbes outperform PRMs in TTS, while being x810 smaller

Idea 2: Use uncertainty to control the reasoning flow[45]



- **Low-uncertainty step** ⇒ continue the default reasoning trajectory *(no test-time scaling)*.
- **High-uncertainty step** ⇒ trigger branching to explore alternatives *(enable test-time scaling)*.
- **(Optional) Stop expanding** highly uncertainty trajectories

[45] Xianzhi Li et al. (2026). "Entropy-Gated Branching for Efficient Test-Time Reasoning". In: *Findings of EACL-2026*.

Impact on compute reduction and performance in TTS[46]

- **Entropy-gated branching** (Xianzhi Li et al. 2026): 31–75% speedup vs. exhaustive TTS; accuracy ↑ by 22.6%
- **MUR** (Hang Yan et al. 2025): up to 60% token reduction, +2-4% avg. accuracy
- **DeepConf** (Yichao Fu et al. 2025a): 43–84% token reduction
- **Entro-duction** (Jinghan Zhang et al. 2025a): 36–94% fewer steps

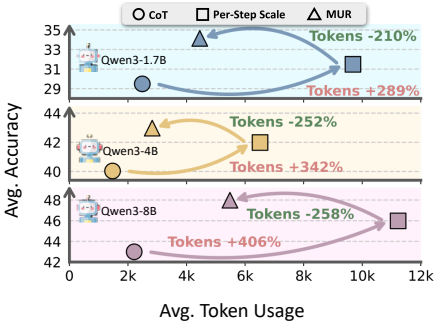


Figure: Token reduction in MUR

[46] Xianzhi Li et al. (2026). "Entropy-Gated Branching for Efficient Test-Time Reasoning". In: *Findings of EACL-2026*; Hang Yan et al. (2025). "Mur: Momentum uncertainty guided reasoning for large language models". In: *arXiv:2507.14958*; Yichao Fu et al. (2025a). "Deep Think with Confidence". In: *Workshop on Eff. Reasoning at NeurIPS 2025.* Spotlight; Jinghan Zhang et al. (2025a). "Entropy-based Exploration Conduction for Multi-step Reasoning". In: *Findings of ACL-2025*.

Idea 3: Early exit during thinking in reasoning LLMs[47]

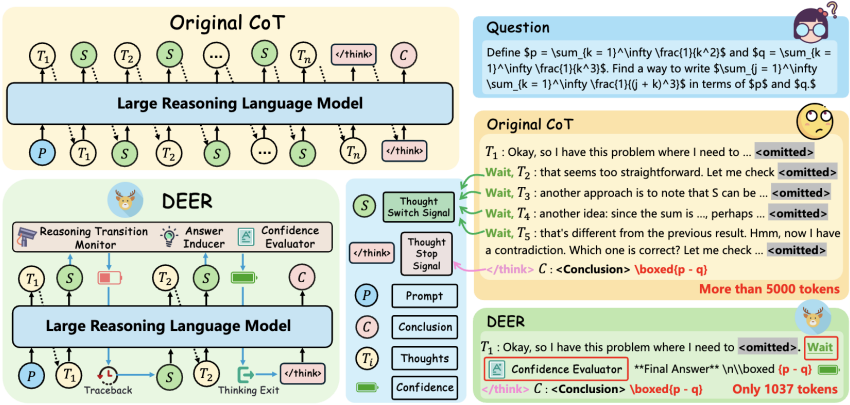


Figure: DEER: dynamic early exit reasoning. Stop thinking if you are already confident

[47] Chenxu Yang et al. (2025a). "Dynamic early exit in reasoning models". In: *arXiv:2504.15895*.

Impact on compute reduction and performance in native thinking

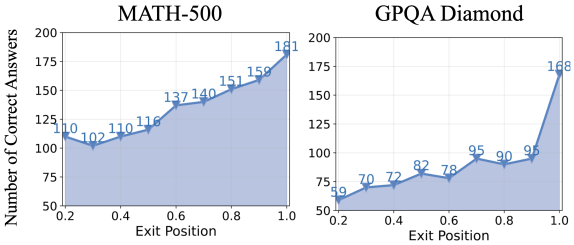


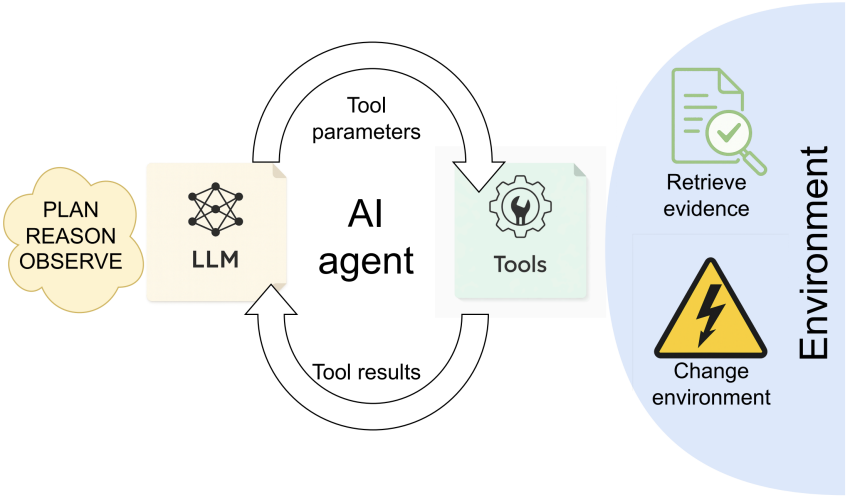
Figure: Number of answers that stay correct with early exit (illustration in DEER)

- DEER[48] ↓ thinking length by 19.1–80.1% while ↑ accuracy by 0.3–5.0%
- Certindex[49] ↓ compute by up to 50%

[48] Chenxu Yang et al. (2025a). "Dynamic early exit in reasoning models". In: *arXiv:2504.15895*.

[49] Yichao Fu et al. (2025b). "Efficiently Scaling LLM Reasoning Programs with Certaindex". In: *NeurIPS-2025*.

Agentic workflow



Idea 4: when uncertain – get more evidence from tools, when confident – skip tools

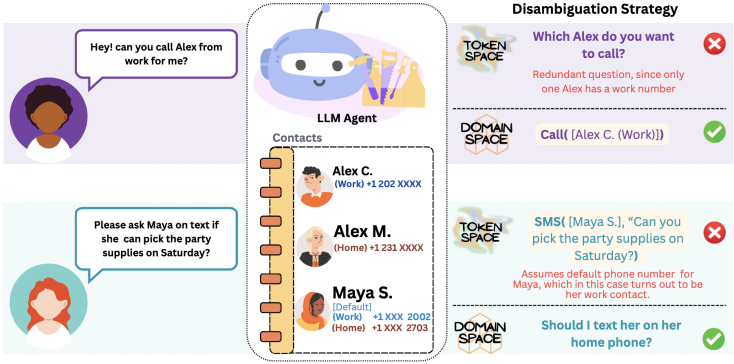
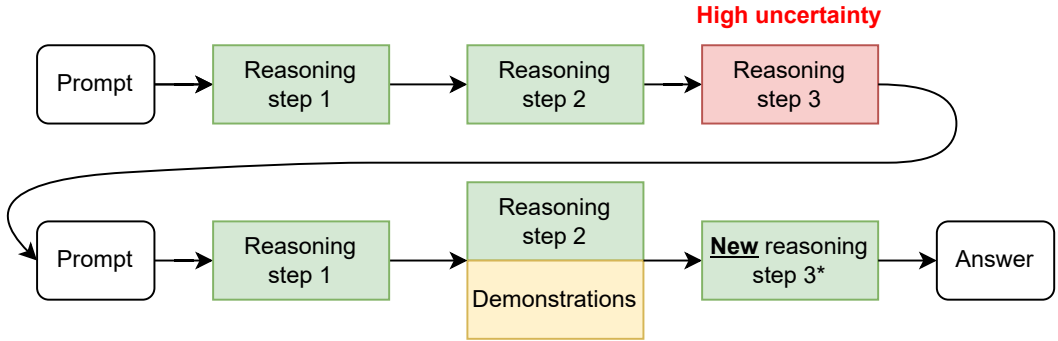


Figure: Visualization from Manan Suri et al. (2025)[50]

[50] Manan Suri et al. (2025). "Structured Uncertainty guided Clarification for LLM Agents". In: *arXiv:2511.08798*.

When uncertain, retrieve more evidence[51]

If you run in an uncertain reasoning step, retrieve a set of relevant demonstrations and restart the reasoning process



Result: 2.7-7.8% ↑ accuracy compared CoT in reasoning (GSM8K, AddSub, AQuA)

[51] Zhangyue Yin et al. (2024). "Reasoning in Flux: Enhancing Large Language Models Reasoning through Uncertainty-aware Adaptive Guidance". In: *ACL 2024*.

Skip tool usage if confident[52]

Core idea – insert an *uncertainty gate* into the Thought–Action–Observation loop:

- Generate an answer with Standard/CoT prompting
- Measure answer uncertainty u; **keep** if $u \leq \tau$, else **activate tool** (Search/Wikipedia)
- Re-assess uncertainty after tool-augmented attempt
- If tool-activation fails to reach an answer within a step budget, fall back to Standard/CoT

Results:

- $\uparrow$ exact match by 4–6 points HotpotQA, StrategyQA, and MMLU
- $\downarrow$ latency compared to baseline by 2-3 times

[52] Jiuzhou Han, Wray Buntine, and Ehsan Shareghi (2024). "Towards Uncertainty-Aware Language Agent". In: *Findings of ACL 2024*.

Impact on efficiency of uncertainty in agentic workflows

- Zhangyue Yin et al. (2024):[53] 2.7-7.8% ↑ accuracy
- Jiuzhou Han et al. (2024):[54] ↑ exact match by 4–6 points; ↓ latency 2-3×
- Hanghui Guo et al. (2025):[55] ↑ F1 by 4.1–6.5 pp; ↓ generation steps by about 2.1×
- Manan Suri et al. (2025):[56] ↑ coverage by about +3–4 pp; 54% ↓ API calls

[53] Zhangyue Yin et al. (2024). "Reasoning in Flux: Enhancing Large Language Models Reasoning through Uncertainty-aware Adaptive Guidance". In: *ACL 2024*.

[54] Jiuzhou Han, Wray Buntine, and Ehsan Shareghi (2024). "Towards Uncertainty-Aware Language Agent". In: *Findings of ACL 2024*.

[55] Hanghui Guo et al. (2025). "DioR: Adaptive Cognitive Detection and Contextual Retrieval Optimization for Dynamic Retrieval-Augmented Generation". In: *ACL-2025*.

[56] Manan Suri et al. (2025). "Structured Uncertainty guided Clarification for LLM Agents". In: *arXiv:2511.08798*.

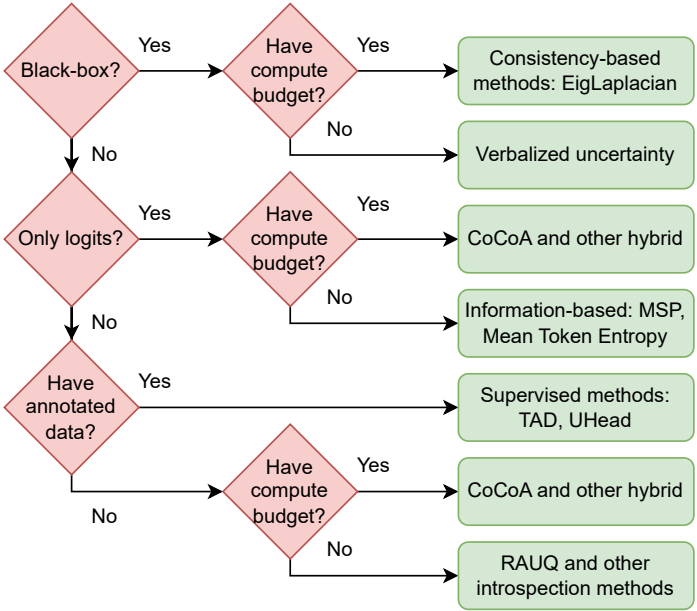
Takeaways

- Uncertainty helps to improve the performance and reduce compute for reasoning and agentic workflows :
  - Uncertainty as a reasoning step scorer
  - Use uncertainty to control the reasoning flow
  - Early exit during thinking in reasoning LLMs
  - Agents: when uncertain – get more evidence from tools, when confident – skip tools
- Why to use specifically uncertainty in reasoning and agents? Because it is usually more efficient compared to PRMs, and other external verifiers
- Even simple compute-cheap methods provide notable impact
- As shown in ReProbes, internal states of LLM provide informative signals about quality of reasoning steps and can be used to improve TTS

9. Conclusion & Future Work

Artem Shelmanov

Summary


Black-box?
Yes
Have compute budget?
Yes
Consistency-based methods: EigLaplacian
No
Verbalized uncertainty
No
Only logits?
Yes
Have compute budget?
Yes
CoCoA and other hybrid
No
Information-based: MSP, Mean Token Entropy
No
Have annotated data?
Yes
Supervised methods: TAD, UHead
No
Have compute budget?
Yes
CoCoA and other hybrid
No
RAUQ and other introspection methods

Our frameworks for research and practice



LM-Polygraph



Test time scaling framework





Uncertainty head

Open research questions

Open research questions

- Researchers have yet to fully explore the potential of uncertainty in reasoning and agentic workflows
  - Need to assess the impact of state-of-the-art UQ methods over baselines

Open research questions

- Researchers have yet to fully explore the potential of uncertainty in reasoning and agentic workflows
  - Need to assess the impact of state-of-the-art UQ methods over baselines
  - Specialized uncertainty and confidence estimation methods for complex reasoning and agentic workflows need deeper investigation, e.g. UQ for structured generation and tool calls

Open research questions

- Researchers have yet to fully explore the potential of uncertainty in reasoning and agentic workflows
  - Need to assess the impact of state-of-the-art UQ methods over baselines
  - Specialized uncertainty and confidence estimation methods for complex reasoning and agentic workflows need deeper investigation, e.g. UQ for structured generation and tool calls
  - Uncertainty can be an additional channel of interaction between agents

Open research questions

- Researchers have yet to fully explore the potential of uncertainty in reasoning and agentic workflows
  - Need to assess the impact of state-of-the-art UQ methods over baselines
  - Specialized uncertainty and confidence estimation methods for complex reasoning and agentic workflows need deeper investigation, e.g. UQ for structured generation and tool calls
  - Uncertainty can be an additional channel of interaction between agents
- How can we adapt UQ methods to new modalities: images, video, audio?

Open research questions

- Researchers have yet to fully explore the potential of uncertainty in reasoning and agentic workflows
  - Need to assess the impact of state-of-the-art UQ methods over baselines
  - Specialized uncertainty and confidence estimation methods for complex reasoning and agentic workflows need deeper investigation, e.g. UQ for structured generation and tool calls
  - Uncertainty can be an additional channel of interaction between agents
- How can we adapt UQ methods to new modalities: images, video, audio?
- Should LLMs be trained to natively produce confidence and uncertainty estimates, or should uncertainty methods be developed on top of the models?

Final takeaways

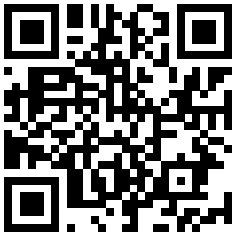
- ML models are always limited due to limited amount of training data or ambiguity of the task
- UQ is a crucial component of ML systems including those based on LLMs
- LM-Polygraph and Uncertainty head frameworks will help you to integrate uncertainty into your code and perform further research
- UQ is a fascinating research direction that might be fruitful not only in terms of safety, but also improve reasoning LLMs and AI agents

Thank you!

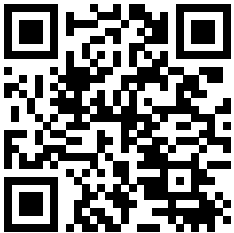
Tutorial:



LM-Polygraph Github:



TACL paper:



10. Q&A

References I

- Azaria, Amos and Tom Mitchell (2023). "The Internal State of an LLM Knows When It's Lying". In: *Findings of EMNLP 2023*.
- Chandu, Khyathi et al. (2025). "CertainlyUncertain: A Benchmark and Metric for Multimodal Epistemic and Aleatoric Awareness". In: *ICLR*.
- Chen, Chao et al. (2024). "INSIDE: LLMs' Internal States Retain the Power of Hallucination Detection". In: *ICLR 2024*.
- Chuang, Yung-Sung et al. (2024). "Lookback Lens: Detecting and Mitigating Contextual Hallucinations in Large Language Models Using Only Attention Maps". In: *EMNLP 2024*.
- Duan, Jinhao et al. (2024). "Shifting Attention to Relevance: Towards the Predictive Uncertainty Quantification of Free-Form Large Language Models". In: *ACL 2024*.
- Fadeeva, Ekaterina et al. (2024). "Fact-Checking the Output of Large Language Models via Token-Level Uncertainty Quantification". In: *Findings of ACL 2024*.

References II

- Fadeeva, Ekaterina et al. (2025). "Don't Throw Away Your Beams: Improving Consistency-based Uncertainties in LLMs via Beam Search". In: *arXiv preprint arXiv:2512.09538*.
- Fang, Yixiong et al. (2025). "Enhancing Vision-Language Model Reliability with Uncertainty-Guided Dropout Decoding". In: *The Thirty-ninth Annual Conference on Neural Information Processing Systems*.
- Farquhar, Sebastian et al. (2024). "Detecting hallucinations in large language models using semantic entropy". In: *Nature*.
- Fomicheva, Marina et al. (2020). "Unsupervised Quality Estimation for Neural Machine Translation". In: *TACL 2020*.
- Fu, Yichao et al. (2025a). "Deep Think with Confidence". In: *Workshop on Eff. Reasoning at NeurIPS 2025*. Spotlight.
- Fu, Yichao et al. (2025b). "Efficiently Scaling LLM Reasoning Programs with Certaindex". In: *NeurIPS-2025*.

References III

- Gautam, Sushant, Michael A. Riegler, and Pål Halvorsen (2025). "HEDGE: Hallucination Estimation via Dense Geometric Entropy for VQA with Vision-Language Models". In: *arXiv preprint arXiv:2511.12693*.
- Guo, Hanghui et al. (2025). "DioR: Adaptive Cognitive Detection and Contextual Retrieval Optimization for Dynamic Retrieval-Augmented Generation". In: *ACL-2025*.
- Han, Jiuzhou, Wray Buntine, and Ehsan Shareghi (2024). "Towards Uncertainty-Aware Language Agent". In: *Findings of ACL 2024*.
- He, Jinwen et al. (2024). "LLM factoscope: Uncovering LLMs' factual discernment through measuring inner states". In: *Findings of ACL 2024*.
- Hüllermeier, Eyke and Willem Waegeman (2021). "Aleatoric and epistemic uncertainty in machine learning: An introduction to concepts and methods". In: *Machine learning*.
- Kang, Zhewei, Xuandong Zhao, and Dawn Song (2025). "Scalable Best-of-N Selection for Large Language Models via Self-Certainty". In: *NeurIPS-2025*.

References IV

- Kotelevskii, Nikita et al. (2025). "From risk to uncertainty: Generating predictive uncertainty measures via Bayesian estimation". In: *ICLR 2025*.
- Kuhn, Lorenz, Yarin Gal, and Sebastian Farquhar (2023). "Semantic Uncertainty: Linguistic Invariances for Uncertainty Estimation in Natural Language Generation". In: *ICLR 2023*.
- Lakshminarayanan, Balaji, Alexander Pritzel, and Charles Blundell (2017). "Simple and scalable predictive uncertainty estimation using deep ensembles". In: *NeurIPS 2017*.
- Lee, Kimin et al. (2018). "A simple unified framework for detecting out-of-distribution samples and adversarial attacks". In: *NeurIPS 2018*.
- Li, Xianzhi et al. (2026). "Entropy-Gated Branching for Efficient Test-Time Reasoning". In: *Findings of EACL-2026*.
- Li, Yifan et al. (2023). "Evaluating Object Hallucination in Large Vision-Language Models". In: *EMNLP*.

References V

- Lin, Chin-Yew (2004). "ROUGE: A Package for Automatic Evaluation of Summaries". In: *Text Summarization Branches Out*.
- Lin, Zhen, Shubhendu Trivedi, and Jimeng Sun (2024). "Generating with Confidence: Uncertainty Quantification for Black-box Large Language Models". In: *TMLR 2024*.
- Ma, Yiran Rex (2025). "Do Androids Question Electric Sheep? A Multi-Agent Cognitive Simulation of Philosophical Reflection on Hybrid Table Reasoning". In: *ACL 2025 (Student Research Workshop)*.
- Malinin, Andrey and Mark J. F. Gales (2021). "Uncertainty Estimation in Autoregressive Structured Prediction". In: *ICLR 2021*.
- Mohri, Christopher and Tatsunori Hashimoto (2024). "Language Models with Conformal Factuality Guarantees". In: *ICML 2024*.
- Ni, Jingwei et al. (2025). *Efficient Test-Time Scaling of Multi-Step Reasoning by Probing Internal States of Large Language Models*. arXiv preprint.

References VI

- Nikitin, Alexander et al. (2024). "Kernel language entropy: Fine-grained uncertainty quantification for LLMs from semantic similarities". In: *NeurIPS 2024* 37.
- Papineni, Kishore et al. (2002). "Bleu: a Method for Automatic Evaluation of Machine Translation". In: *ACL 2002*.
- Qiu, Xin and Risto Miikkulainen (2024). "Semantic Density: Uncertainty Quantification for Large Language Models through Confidence Measurement in Semantic Space". In: *NeurIPS 2024*.
- Rei, Ricardo et al. (2020). "COMET: A Neural Framework for MT Evaluation". In: *EMNLP 2020*.
- Reimers, Nils and Iryna Gurevych (2019). "Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks". In: *EMNLP-IJCNLP 2019*.
- Ren, Jie et al. (2023). "Out-of-Distribution Detection and Selective Generation for Conditional Language Models". In: *ICLR 2023*.

References VII

- Seo, Hoigi et al. (2025). "On Epistemic Uncertainty of Visual Tokens for Object Hallucinations in Large Vision-Language Models". In: *NeurIPS 2025*.
- Shelmanov, Artem et al. (2025). "A Head to Predict and a Head to Question: Pre-trained Uncertainty Quantification Heads for Hallucination Detection in LLM Outputs". In: *EMNLP 2025*.
- Slobodkin, Aviv et al. (Dec. 2023). "The Curious Case of Hallucinatory (Un)answerability: Finding Truths in the Hidden States of Over-Confident Large Language Models". In: *EMNLP 2023*.
- Sriramanan, Gaurang et al. (2024b). "LLM-Check: Investigating Detection of Hallucinations in Large Language Models". In: *NeurIPS 2024*.
- — (2024a). "LLM-Check: Investigating Detection of Hallucinations in Large Language Models". In: *NeurIPS 2024*. Vol. 37.
- Su, Weihang et al. (Aug. 2024). "Unsupervised Real-Time Hallucination Detection based on the Internal States of Large Language Models". In: *Findings of ACL 2024*.

References VIII

- Suri, Manan et al. (2025). "Structured Uncertainty guided Clarification for LLM Agents". In: *arXiv:2511.08798*.
- Tian, Katherine et al. (2023). "Just Ask for Calibration: Strategies for Eliciting Calibrated Confidence Scores from Language Models Fine-Tuned with Human Feedback". In: *EMNLP 2023*.
- Vashurin, Roman et al. (2025a). "Benchmarking uncertainty quantification methods for large language models with lm-polygraph". In: *TACL 2025*.
- Vashurin, Roman et al. (2025b). "UNCERTAINTY-LINE: Length-Invariant Estimation of Uncertainty for Large Language Models". In: *EMNLP 2025*.
- Vashurin, Roman et al. (2025c). "Uncertainty Quantification for LLMs through Minimum Bayes Risk: Bridging Confidence and Consistency". In: *NeurIPS 2025*.
- Vazhentsev, Artem et al. (2023). "Hybrid Uncertainty Quantification for Selective Text Classification in Ambiguous Tasks". In: *ACL 2023*.

References IX

- Vazhentsev, Artem et al. (2025a). "Token-Level Density-Based Uncertainty Quantification Methods for Eliciting Truthfulness of Large Language Models". In: *NAACL 2025*.
- Vazhentsev, Artem et al. (2025b). "Uncertainty-Aware Attention Heads: Efficient Unsupervised Uncertainty Quantification for LLMs". In: *arXiv preprint arXiv:2505.20045*.
- Vazhentsev, Artem et al. (2025c). "Unconditional Truthfulness: Learning Conditional Dependency for Uncertainty Quantification of Large Language Models". In: *EMNLP 2025*.
- Yan, Hang et al. (2025). "Mur: Momentum uncertainty guided reasoning for large language models". In: *arXiv:2507.14958*.
- Yang, Chenxu et al. (2025a). "Dynamic early exit in reasoning models". In: *arXiv:2504.15895*.

References X

- Yang, Ziran et al. (2025b). "Understanding the Sources of Uncertainty for Large Language and Multimodal Models". In: *ICLR Workshop: QUESTION*.
- Yin, Zhangyue et al. (2024). "Reasoning in Flux: Enhancing Large Language Models Reasoning through Uncertainty-aware Adaptive Guidance". In: *ACL 2024*.
- Yoo, KiYoon et al. (2022). "Detection of Adversarial Examples in Text Classification: Benchmark and Baseline via Robust Density Estimation". In: *Findings of ACL 2022*.
- Zha, Yuheng et al. (2023). "AlignScore: Evaluating Factual Consistency with A Unified Alignment Function". In: *ACL 2023*.
- Zhang, Caiqi et al. (2024). "LUQ: Long-text Uncertainty Quantification for LLMs". In: *EMNLP 2024*.
- Zhang, Jinghan et al. (2025a). "Entropy-based Exploration Conduction for Multi-step Reasoning". In: *Findings of ACL-2025*.
- Zhang, Tianhang et al. (2023). "Enhancing Uncertainty-Based Hallucination Detection with Stronger Focus". In: *EMNLP 2023*.

References XI

- Zhang, Tianyi et al. (2020). "BERTScore: Evaluating Text Generation with BERT". In: *8th International Conference on Learning Representations, ICLR 2020*.
- Zhang, Zhenliang et al. (2025b). "ICR Probe: Tracking Hidden State Dynamics for Reliable Hallucination Detection in LLMs". In: *ACL 2025*.